



Robust ship-to-grid scheduling problem: A contextual joint estimation-prediction-optimization paradigm

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ARTICLE INFO

Keywords:

Contextual optimization
Joint estimation–prediction–optimization
framework
Ship scheduling
Ship-to-grid

ABSTRACT

As pollution and climate change intensify, mitigating emissions from the shipping sector has become a global priority for protecting public health and biodiversity. All-electric ships (AESs) have consequently emerged as a promising low-carbon alternative, supported by increasingly stringent policy mandates. In parallel, Ship-to-Grid (S2G) technology enables bidirectional energy exchange between AESs and port power grids, effectively transforming vessels into mobile energy storage units. This dual operational role, however, fundamentally complicates traditional ship scheduling problem. In this study, we develop a dynamic AES scheduling model on a spatial-temporal network that jointly determines freight routing, vessel relocation, and coordinated charging/discharging decisions. To capture context-dependent uncertainty in freight demand, we propose a contextual joint estimation-prediction-optimization framework that establishes a unified data-to-decision paradigm by embedding estimation and prediction procedures directly into the uncertainty set. By adopting linear decision rules together with a lifted uncertainty set construction, we show that the resulting model admits a tractable reformulation. We validate the model using both simulation studies and real-world instances from the ports in the North Sea region of Europe. Results show that our framework delivers superior out-of-sample performance in both average and risk-averse settings. Moreover, comparative experiments indicate that incorporating S2G operations provides additional revenue improvements, particularly during periods of pronounced electricity price fluctuations.

1. Introduction

Human society is increasingly grappling with the devastating impacts of environmental degradation. Unsustainable industrial practices and excessive consumption are fueling air pollution, poisoning our waters, and disrupting ecosystems, all of which threaten biodiversity and the well-being of public health. The shipping industry stands as a major contributor in this crisis, heavily relying on fossil fuels, and producing nearly 3% of global CO₂ emissions, more than the whole emissions of advanced economies like Germany or Japan (IMO, 2023). Without decisive action, emissions from this sector could soar by as much as 50% by 2050, highlighting the urgent need for low-carbon technologies in maritime transport to ensure a sustainable future.

As the maritime sector faces mounting pressure to mitigate its environmental externalities, electrification has emerged as a pivotal path toward decarbonization. By replacing fossil-fuel propulsion, electric vessels eliminate direct emissions of CO₂, NO_x, and SO_x, while also cutting underwater noise pollution by 40–50 dB, an essential improvement for the protection of marine ecosystems (DNV, 2024). Policy frameworks are accelerating this transformation, with the IMO's 2023 GHG Strategy mandating that 90% of major

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<https://doi.org/10.1016/j.trb.2026.103569>

Received 17 December 2025; Received in revised form 21 July 2026; Accepted 31 July 2026

Available online 15 August 2026

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ports will be shore-power compatible by 2030 (IMO, 2023), and the UN-endorsed Clydebank Declaration establishing zero-emission corridors to prioritize electric ships across 24 nations (Department for Transport, 2021). As a result, maritime electrification has progressed from experimental pilots to commercially viable deployments, with all-electric ships (AESs) increasingly becoming a mainstream solution for reducing the environmental footprint.

Building on the growing adoption of AESs in green ports, Ship-to-Grid (S2G) technology offers a compelling opportunity to further enhance energy efficiency and accelerate the integration of renewable energy. By enabling bidirectional energy flow between berthed AESs and the port's power grid, S2G turns AESs into mobile energy storage units, allowing them to not only draw power when needed but also feed surplus energy back into the grid, fostering a dynamic and sustainable energy ecosystem. S2G brings substantial benefits to both ports and shipping companies. For ports, it facilitates flexible freight management, helping stabilize the power grid while also providing a reliable backup power source during emergencies. For shipping companies, it opens up new revenue streams by enabling the sale of surplus energy back to the grid. As a result, government policy initiatives are accelerating the adoption of S2G, with examples such as the EU's 2030 infrastructure mandate (EU, 2021) and China's financial incentives for retrofits (MOT, 2024). These initiatives highlight S2G's potential to strengthen port resilience and contribute to the goal of maritime decarbonization.

The integration of S2G technology substantially increases the complexity of the conventional ship scheduling problem. In addition to planning freight transportation and vessel relocation, shipping operators must also coordinate AESs' charging and discharging operations to effectively interact with the port's power grid. For example, during periods of high electricity demand, AESs can discharge stored energy back into the grid to stabilize grid performance and enhance economic returns, whereas in off-peak periods, they may prioritize recharging to ensure adequate power for subsequent voyages. Within this framework, AESs assume a dual role, as both transportation carriers and mobile energy storage units capable of storing and supplying electricity. This dual role fundamentally reshapes the conventional ship scheduling paradigm and highlights the need for more advanced and integrated decision-making frameworks.

Another critical challenge in AESs scheduling arises from the uncertainty of freight demand. Unlike conventional ships that serve broad and relatively stable cargo streams, where freight demand is typically treated as static and stable, AESs concentrate on a narrower range of time-sensitive and high-value commodities, such as e-commerce parcels, medical supplies, and perishables (Ivanov, 2022). Accordingly, freight demand of the AESs scheduling model is heavily influenced by contextual information such as weather conditions, and hinterland characteristics, rendering it inherently context-dependent. For example, fluctuations in weather conditions, such as extreme temperatures or storms, can impact the operating conditions and service availability of AESs, thereby affecting freight demand. Meanwhile, the characteristics of the hinterland, including industrial structure, regional economic activity, and short-term consumption patterns, can also directly influence demand. Failing to account for such contextual information prevents the model from accurately capturing freight demand, which may lead to biased forecasts and suboptimal scheduling outcomes. Therefore, addressing the AESs scheduling problem necessitates the integration of contextual information to better predict uncertain freight demand and generate more reliable and effective operational decisions.

In this study, we introduce a novel dynamic AESs scheduling problem by integrating S2G technology, explicitly considering the context-dependent uncertainty in freight demand. To the best of our knowledge, this is the first attempt to explore the AESs operations within the S2G framework. We focus on the following two research questions: (i) How can we formulate an optimization model for the dynamic AESs scheduling problem that simultaneously addresses decision-making for AESs chartering, freighting, relocating, and charging/discharging operations? (ii) How can we construct a reliable uncertainty set for context-dependent uncertainty while maintaining model tractability and optimality? By solving these research questions, we summarize our contributions as follows.

- (1) **Enriching the ship scheduling problem.** After constructing spatial-temporal-SoC network, we propose a dynamic AESs scheduling model for shipping company to produce operational scheduling scheme, which introduces the following new aspects:
 - We incorporate S2G operations into the dynamic AESs scheduling problem, meaning that beyond freight routing and AESs relocation, the model must also coordinate AESs' charging and discharging activities to interact effectively with the port's power grid.
 - We explicitly account for context-dependent uncertainty in AES freight demand, whose variation can be partially explained by observable contextual information such as weather conditions and port throughput.
- (2) **Extending the existing methodology.** We propose a contextual joint estimation-prediction-optimization (JEPO) framework, which advances the methodology in the following aspects:
 - Beyond the classical norm-based metric, we propose a contextual disparity metric that embeds a deviation matrix, thereby allowing deviations to be evaluated in a more principled manner that reflects dimensional heterogeneity.
 - In contrast to the sequential process of estimation, prediction, and optimization, the proposed JEPO framework offers a unified data-to-decision paradigm by embedding the estimation and prediction procedures directly into the uncertainty set of the optimization model.
 - We propose a data-driven calibration approach for the JEPO uncertainty set by identifying the minimum volume confidence ball without compromising the confidence level, thereby mitigating the conservatism of the robust solutions.
 - Using a linear decision rule approximation and a lifted uncertainty set construction, we demonstrate that the contextual JEPO model admits a tractable second-order cone program (SOCP) formulation.
- (3) **Demonstrating superior performance in numerical experiments.** We validate the practical effectiveness of our approach using both simulation studies and real-world instances from the ports in the North Sea region of Europe. Compared with two benchmarks, our framework delivers superior out-of-sample performance in both average and risk-averse settings, achieving revenue gains of

12.69% and 17.78%, respectively. Additionally, comparative experiments indicate that incorporating S2G operations provides additional revenue improvements, particularly during periods of pronounced electricity price fluctuations.

The remainder of this paper is organized as follows. [Section 2](#) reviews the related literature. [Section 3](#) develops a mathematical model for the dynamic AESs scheduling problem. [Section 4](#) handles the context-dependent uncertainty and construct a JEPO uncertainty set. [Section 5](#) focuses on the tractability of the model. In [Section 6](#), we validate the effectiveness of our methodology through a numerical experiment. Finally, a brief conclusion and managerial insights are provided in [Section 7](#).

Notation. We use boldface uppercase and lowercase letters to denote matrices, e.g., $\mathbf{H}$, and vectors, e.g., $\mathbf{x}$, respectively. The inverse of matrix $\mathbf{H}$ is denoted by $\mathbf{H}^{-1}$. We denote by $\mathbb{R}^{I^2T}$ the I^2T -dimensional Euclidean space. For a positive integer I , we use $[I] := \{1, 2, \dots, I\}$ to denote the corresponding index set. We further denote by $\mathbb{S}_{++}^{I^2T}$ the set of all positive definite matrices in $\mathbb{R}^{I^2T \times I^2T}$, namely, $\mathbb{S}_{++}^{I^2T} := \{\mathbf{H} \in \mathbb{R}^{I^2T \times I^2T} \mid \mathbf{H} > 0\}$. Random variables are denoted by a tilde; for example, $\tilde{d} \sim \mathbb{P}$, where $\mathbb{P} \in \mathcal{P}_0(D)$, and $\mathcal{P}_0(D)$ denotes the set of all probability distributions supported on $D \subseteq \mathbb{R}^{I^2T}$. We use $\mathbb{E}_{\mathbb{P}}[\cdot]$ to denote expectation under distribution $\mathbb{P}$. Finally, for any scalar a , we define its nonnegative part as $[a]^+ := \max\{a, 0\}$.

2. Literature review

In this section, we review the related literature. Specifically, [Sections 2.1](#) and [2.2](#) examine the literature on the standard ship scheduling problem and maritime electrification, respectively, while [Section 2.3](#) provides an overview of methodological studies on optimization under uncertainty.

2.1. Standard ship scheduling problem

In maritime logistics, ship scheduling plays an indispensable and pivotal role, involving the optimization of fleet deployment and freight routing under multiple sources of uncertainty. Early works, such as [Christiansen et al. \(2004\)](#), provide a systematic review of ship routing and scheduling problems across strategic, tactical, and operational levels. They separately discuss two types of shipping operations: tramp and liner shipping, which exhibit distinct differences in their scheduling strategies. Specifically, tramp shipping is characterized by high operational flexibility. Vessels are deployed according to cargo availability rather than fixed routes or schedules, allowing shipping companies to maximize profits by dynamically optimizing scheduling decisions. In contrast, liner shipping follows fixed routes and schedules and its operations focusing on strategic network design, fleet sizing, and fleet deployment.

Tramp shipping scheduling primarily aims to optimize routing and bunkering decisions to maximize profits under uncertain conditions. [Vilhelmsen et al. \(2014\)](#) examine the integration of bunkering decisions with routing, suggesting that incorporating bunkering considerations during the routing phase enhances cargo allocation and efficiency, particularly in the face of fluctuating fuel prices. Building on this, [Omholt-Jensen et al. \(2025\)](#) model the tramp ship routing and scheduling problem with bunker optimization as a two-stage stochastic programming problem. They propose that integrating bunkering decisions with routing can increase profits by adapting to uncertainties in demand and fuel costs. Additionally, [Norstad et al. \(2011\)](#) explore speed optimization, arguing that treating speed as a decision variable rather than a fixed parameter which improves operational efficiency by more effectively managing fuel consumption. [Wen et al. \(2016\)](#) extend this concept by jointly optimizing routing and speed, suggesting that this combined approach enhances fleet profitability and overall operational performance.

Liner shipping scheduling is fundamental to ensuring service quality and operational efficiency. Early research primarily employs deterministic models with fixed sailing speeds, port times, and demand, aiming to optimize costs or profits ([Meng et al., 2010](#); [Qi et al., 2012](#)). However, such models often neglect operational uncertainties, including port congestion, handling delays, and adverse weather conditions. More recent studies incorporate stochastic elements into scheduling frameworks by applying fuzzy logic, chance-constrained programming, and simulation-based optimization methods ([Chuang et al., 2010](#); [Wang et al., 2012](#)). To enhance schedule reliability in the face of disruptions, researchers increasingly examine recovery strategies such as speed adjustments, port skipping, and container diversion ([Song et al., 2015](#); [Gürel et al., 2019](#)).

While existing tramp and liner scheduling studies focus on conventional fuel-powered ships, the scheduling of AESs remains underexplored. The distinct operational features of AESs, particularly the need to coordinate charging/discharging operations with cargo freighting and ship relocating, call for specialized AESs scheduling strategies. To bridge this gap, we conceptualize a dynamic AESs scheduling model in the context of S2G, which explicitly accounts for the context-dependent uncertainty in freight demand.

2.2. Maritime electrification

This study is also related to maritime electrification, which plays a vital role in reducing emissions and enhancing energy efficiency within the shipping industry. The adoption of S2G technology has transformed the energy management of conventional vessels, thereby influencing the optimization and scheduling of AESs. To lay the foundation for addressing the AES scheduling problem, this section focuses on AESs and S2G, which are essential components in advancing maritime electrification.

Efficient voyage scheduling becomes a crucial factor in optimizing the operations of AESs. Early research by [Kanellos et al. \(2014\)](#) focuses on coordinating voyage scheduling with power dispatch in hybrid power systems. They develop a dynamic programming model that simultaneously optimizes AES speed and onboard power generation to reduce overall energy consumption. Recognizing that speed optimization alone may limit operational flexibility, [Wen et al. \(2020\)](#) extend this work by introducing a joint optimization approach that integrates voyage planning with the sizing of onboard energy storage systems, aiming to enhance both economic

efficiency and renewable energy utilization. As scheduling objectives become increasingly complex, [Shang et al. \(2016\)](#) propose a bi-objective model that considers both operational costs and greenhouse gas emissions, solved using NSGA-II to derive Pareto-optimal scheduling strategies. Building on these advances, [Hein et al. \(2021\)](#) develop a multi-objective framework for voyage planning and energy management in AESs equipped with hybrid energy storage systems, striving to balance environmental sustainability and cost efficiency under practical maritime operational constraints. However, existing scheduling literature on AESs primarily emphasizes energy-related aspects, often overlooking the scheduling challenges that arise in shipping networks under uncertain freight demand.

Simultaneously, the integration of S2G technology has attracted growing attention in the context of AESs. This technology enables AESs to not only draw power from the grid but also feed power back when required. Although research on S2G is still in its early stages and existing studies primarily focus on technical aspects ([Reed et al., 2012](#); [Vlachokostas et al., 2019](#); [Badakhshan et al., 2022](#)), related research on Vehicle-to-Grid (V2G) operation management provides valuable insights for our work. [Peterson et al. \(2010\)](#) examine the economic implications of using electric vehicle batteries for grid electricity storage, finding that while battery degradation impacts profitability, the potential for energy arbitrage remains significant. [Zhang et al. \(2021\)](#) further explore V2G integration in electric vehicle sharing systems, demonstrating that V2G increases profitability and service quality.

Although S2G and V2G both rely on bidirectional charging technology, they differ substantially in their operational settings. In V2G systems, vehicle mobility is determined by individual users, who decide when and where their EVs travel. As a result, the operator plays a largely passive role and can only predict vehicle availability based on travel behavior and participation patterns. Accordingly, V2G typically operates in a decentralized manner, where the timing and location of charging and discharging cannot be directly controlled by the operator. Instead, the operator usually relies on incentive mechanisms, such as financial compensation, to encourage EV owners to participate in charging or discharging activities at desired times. By contrast, S2G systems are centrally managed by the shipping company, which directly determines the operations of AESs. This centralized structure allows the operator to actively coordinate freight transportation as well as charging and discharging decisions. In particular, the operator can jointly optimize routing, freight scheduling, and energy transactions in response to variations in freight demand and electricity prices. Therefore, unlike V2G, S2G does not rely on voluntary participation through incentives, but instead is implemented through centralized operational decisions made by the company. However, research on S2G operation management strategies remains limited. To address this gap, our study proposes a scheduling framework for AESs that integrates S2G technology. The framework aims to enhance operational efficiency and optimize revenue generation, particularly in the context of uncertainties associated with freight demand.

2.3. Optimization under uncertainty

Conventionally, uncertainty in optimization problems has been addressed primarily through stochastic programming (SP) and robust optimization (RO). While both have formed the foundation of uncertainty modeling, they suffer from critical limitations. Specifically, SP relies on the precise characterization of the probability distribution associated with uncertain parameters—an assumption that is often unrealistic in many practical scenarios. On the other hand, RO completely excludes probabilistic information and instead focuses on worst-case outcomes, which tends to produce overly cautious decisions. To mitigate these shortcomings, distributionally robust optimization (DRO) has emerged as a compelling alternative ([Wang et al., 2020](#)). Despite their theoretical appeal, constructing such sets using empirical data and domain-specific knowledge remains a substantial challenge. In response, [Long et al. \(2023\)](#) introduce the robust satisficing framework, which eliminates the need for specifying a distributional set altogether. More recently, [Sim et al. \(2024\)](#) further advance this framework by incorporating predictive models and decision-making mechanisms, enhancing its practicality across diverse uncertain environments.

Contextual robust optimization has recently attracted significant attention for its ability to incorporate covariate information into decision-making under uncertainty ([Mišić and Perakis, 2020](#)). Within this paradigm, three major frameworks have emerged for learning decision policies from data. The decision rule framework, first introduced to operations research by [Liyanage and Shanthikumar \(2005\)](#) and later popularized by [Ban and Rudin \(2019\)](#), directly parameterizes decision policies and optimizes their parameters through empirical risk minimization. The sequential learning and optimization (SLO) framework, formalized by [Bertsimas and Kallus \(2020\)](#), follows a two-step procedure: it first estimates the conditional distribution of uncertain parameters given covariates, and then solves the resulting stochastic optimization problem. Acknowledging the potential mismatch between predictive accuracy and decision quality in SLO, the integrated learning and optimization (ILO) framework jointly trains the predictive model and decision policy by directly minimizing the downstream decision loss—an idea originating in [Bengio \(1997\)](#) and later revitalized under the paradigms of decision-focused learning and smart predict-then-optimize ([Donti et al., 2017](#); [Elmachtoub and Grigas, 2022](#)).

While these above research focus on learning decision policies, a critical yet often overlooked aspect is how to construct the uncertainty set that effectively reflect the data distribution. Traditional methods typically assume that uncertainty set parameters are known or set through empirical means and covariance. However, they usually lack a systematic way to estimate these parameters based on the data distribution and the chosen norm, which can lead to inaccurate uncertainty sets, resulting in solutions that may be overly conservative or insufficiently stable. One common approach uses asymptotic behavior, such as the central limit theorem ([Bandi and Bertsimas, 2012](#)), which works well with large sample sizes but can be overly conservative in small samples sizes, especially with skewed or heavy-tailed distributions. To reduce this conservatism, methods like divergence functionals ([Ben-Tal et al., 2013](#)) and hypothesis testing ([Bertsimas et al., 2018](#)) have been incorporated. [Hong et al. \(2021\)](#) propose a statistical framework that learns uncertainty sets using geometric shapes and data-splitting validation. However, these methods typically require large, high-quality datasets for reliable performance. To address this challenge, [Lou et al. \(2024\)](#) focus on developing an estimation method when only a limited number of samples are available. Specifically, they construct norm-

specific uncertainty sets without relying on restrictive distributional assumptions, while ensuring stable performance in small-sample scenarios.

In this study, we develop a unified contextual JEPO framework that joint parameter estimation, demand prediction, and decision optimization by integrating contextual information. The framework aims to identify optimal decisions that satisfy specific operational revenue target, while explicitly balancing robustness and conservatism.

3. Dynamic AESs scheduling problem

In this section, we focus on the dynamic AESs scheduling problem integrated with S2G charging and discharging operations. Section 3.1 introduces the problem description, followed by Section 3.2, which constructs the time-space-SoC network for dynamic AESs scheduling. Finally, Section 3.3 presents the corresponding mathematical formulation.

3.1. Problem description

In the context of maritime electrification, we consider a shipping company operating a fleet of AESs that provide freight services across a set of ports denoted by $[I] = \{1, \dots, I\}$. The planning horizon is divided into T discrete time periods, represented by $[T] = \{1, \dots, T\}$. At the beginning of the planning horizon, the company determines a strategic decision $x_i \in \mathbb{R}_+$, indicating the number of AESs to be chartered at port $i \in [I]$, subject to the overall budget constraint M .

In contrast to conventional fuel-powered vessels, the feasible sailing range of AESs depends on their remaining battery capacity, quantified by the *State of Charge (SoC)*, which is discretized into B levels denoted by $[B] = \{1, \dots, B\}$. Without loss of generality, we assume that newly chartered AESs begin the planning horizon fully charged, i.e., with an SoC level of B in the first period. Additionally, in the dynamic scheduling of AESs, we focus on the integration of S2G operations, including both battery charging and discharging. Because electricity prices vary across periods, the shipping company can profit by charging during low-price periods and discharging during high-price periods. Formally, we denote the unit electricity prices for charging and discharging at period $t \in [T]$ by p_t^c and p_t^d , respectively.

Within this problem context, the chartered AESs are used to fulfill the uncertain freight demand across the planning horizon. Specifically, freight demand typically varies across ports and time periods due to factors such as seasonality and hinterland economic indicators, rendering it contextual-based uncertain as illustrated in Fig. C.1. Let $\tilde{d}_{i,j,t}$ denote the uncertain freight demand transported from port $i \in [I]$ to port $j \in [I]$ at period $t \in [T]$. We define $\tilde{\mathbf{d}} = (\tilde{d}_{i,j,t})_{i,j \in [I], t \in [T]}$ as the random vector of freight demands, whose support set is $\mathcal{D} \subseteq \mathbb{R}^{I^2 T}$. In practice, although the true freight demand $\tilde{\mathbf{d}}$ is unknown at the time of decision making, certain contextual information that affects demand is observable. We denote this information by a vector $\mathbf{v} \in \mathbb{R}^K$, which may include factors such as weather conditions and port throughput. Ignoring such contextual information and relying only on unconditional demand distributions may lead to systematic prediction bias and, consequently, suboptimal scheduling decisions. In the AESs scheduling setting, a context-agnostic model cannot distinguish high-demand scenarios associated with favorable weather or strong port throughput from low-demand scenarios induced by adverse operating conditions. As a result, it may either over-charter AESs, thereby incurring unnecessary idle and charging costs, or under-charter AESs, thereby causing unmet freight demand and lost revenue.

Nevertheless, freight demand across ports is typically unbalanced. Some ports are primarily exporters with large outbound cargo but little inbound cargo, while others are mainly importers. Consequently, AESs may need to relocate after completing the previous cargo transportation to meet new freight demand.

In the context of uncertain freight demand, shipping company operators face the challenge of determining when and where to transport cargo, charge or discharge batteries of AESs, and relocate to meet new freight demand, with the overall objective of minimizing the total net cost. Unlike conventional ship scheduling problems, on the one hand, our study incorporates additional charging and discharging decisions; on the other hand, the revenue of the AESs scheduling problem derives from two components which include cargo transportation revenue and discharging revenue. Due to the complexity of AESs scheduling problem, which involves three dimensions, time period, port, and SoC level, it requires implementation on a spatial-temporal-SoC network, as introduced in Section 3.2.

3.2. The spatial-temporal-SoC network

In this subsection, we construct a spatial-temporal-SoC network for the proposed dynamic AESs scheduling problem. Formally, the network is defined as $\mathcal{G} = (\mathcal{N}, \mathcal{A})$, where $\mathcal{N}$ and $\mathcal{A}$ denote the sets of nodes and arcs, respectively. Each node $n_{i,t,b} \in \mathcal{N}$ represents the system state of an AES identified by the triplet (i, t, b) , indicating its location at port $i \in [I]$, time period $t \in [T]$, and State-of-Charge (SoC) level $b \in [B]$. Arcs are introduced between node pairs to represent all feasible transitions of AESs across these states. The network includes five types of arcs, each capturing a specific class of AES movements. Fig. 1 illustrates an example of the five arc types, which we describe in detail below.

1. **Freighting arcs:** The freighting arcs represent the transitions of AESs transporting cargo from an origin port $i \in [I]$ to a destination port $j \in [I]$. The sailing time between ports i and j is $l_{i,j}$ periods, and the corresponding SoC consumption is $e^F l_{i,j}$, where e^F denotes the per-period SoC consumption. Let $\mathcal{A}^{\text{frgt}} \subseteq \mathcal{A}$ denote the set of freighting arcs. Then, an arc $(n_{i,t,b}, n_{j,t+l_{i,j}, b-e^F l_{i,j}})$ belongs to $\mathcal{A}^{\text{frgt}}$ if $t + l_{i,j} \in [T]$ and $b - e^F l_{i,j} \in [B]$. For convenience, we further define $\mathcal{D}_{i,j,t}^{\text{frgt}} = \{(n_{i,t,b}, n_{j,t+l_{i,j}, b-e^F l_{i,j}}) \in \mathcal{A}^{\text{frgt}} : b - e^F l_{i,j} \in [B]\}$, which represents the set of freighting arcs that depart from port $i \in [I]$ to port $j \in [I]$ at period $t \in [T]$.

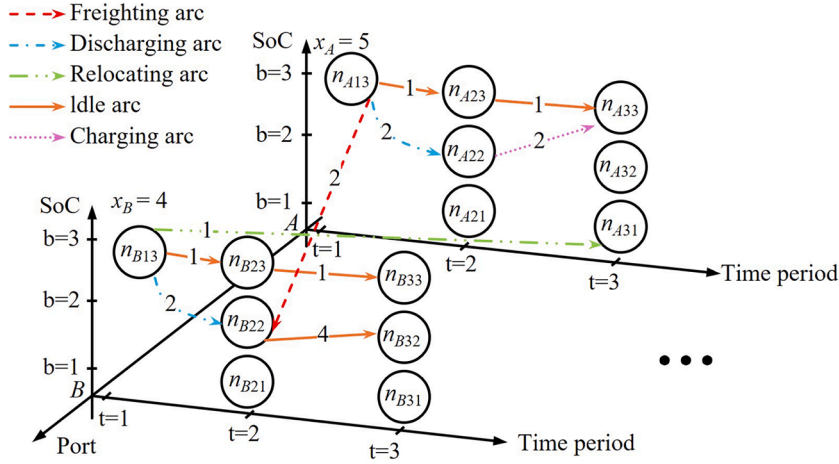


Fig. 1. An example of the spatial-temporal-SoC shipping network.

2. **Relocating arcs:** Because freight demand is often imbalanced across ports, AESs may need to be relocated among the I ports. We denote by $\mathcal{A}^{\text{relo}} \subseteq \mathcal{A}$ the set of relocating arcs, where each arc $(n_{i,t,b}, n_{j,t+1,b+e^R}) \in \mathcal{A}^{\text{relo}}$ is defined whenever $t+1 \in [T]$ and $b+e^R \in [B]$. Here, e^R denotes the relocating cost per unit flow per time period.
3. **Charging arcs:** Upon arriving at port $i \in [I]$, AESs may charge the batteries, gaining e^C SoC levels per period. We denote the set of charging arcs as $\mathcal{A}^{\text{chr}} \subseteq \mathcal{A}$ where each arc $(n_{i,t,b}, n_{i,t+1,b+e^C}) \in \mathcal{A}^{\text{chr}}$ is defined whenever $t+1 \in [T]$ and $b+e^C \in [B]$, with a charging cost of $p_i^c + c^{\text{deg}}$ per unit flow, where c^{deg} represents the degradation cost per SoC.
4. **Discharging arcs:** During high-price periods, AESs may discharge surplus energy back to the power grid, releasing e^D SoC levels per period. We denote the set of discharging arcs as $\mathcal{A}^{\text{dsch}} \subseteq \mathcal{A}$ where each arc $(n_{i,t,b}, n_{i,t+1,b-e^D}) \in \mathcal{A}^{\text{dsch}}$ is defined whenever $t+1 \in [T]$ and $b-e^D \in [B]$, with a discharging revenue of $p_i^s - c^{\text{deg}}$ per unit flow.
5. **Idle arcs:** Under certain cases, AESs may stay docked at port $i \in [I]$ without charging or discharging, either awaiting upcoming tasks or undergoing maintenance. We denote the set of idle arcs as $\mathcal{A}^{\text{idle}} \subseteq \mathcal{A}$ where each arc $(n_{i,t,b}, n_{i,t+1,b}) \in \mathcal{A}^{\text{idle}}$ is defined whenever $t+1 \in [T]$, with a idle cost of c_i^{idle} per unit flow, where c_i^{idle} represents the cost of an AES idling at port $i \in [I]$ per time period.

For simplicity, we denote $\mathcal{A}^{\text{else}} := \mathcal{A}^{\text{relo}} \cup \mathcal{A}^{\text{idle}} \cup \mathcal{A}^{\text{chr}} \cup \mathcal{A}^{\text{dsch}}$. To represent the number of AESs on each type of arc, we define the flow decision variables y_a^{frgt} , $a \in \mathcal{A}^{\text{frgt}}$ and y_a^{else} , $a \in \mathcal{A}^{\text{else}}$ corresponding to the flows on freight arcs, relocating arcs, idle arcs, charging arcs, and discharging arcs, respectively. Fig. 1 illustrates an example of the spatial-temporal-SoC network with two ports, i.e. A and B, three time periods, i.e. $[T] = \{1, 2, 3\}$, and three SoC levels, i.e. $[B] = \{1, 2, 3\}$. The numbers alongside different types of arcs represent the solution for freighting, relocating, idle, charging, and discharging across various ports and time periods.

3.3. Dynamic AESs scheduling model

In this subsection, we introduce a scheduling model for AESs that integrates both charging and discharging mechanisms. Based on the notation summary provided in Table A.1, the model aims to minimize the net cost within a dynamic system represented by the spatial-temporal-SoC network $\mathcal{G}$. Furthermore, we consider the influence of uncertainty in freight demand on the multi-stage problem.

Net cost objective. The net cost objective is a key goal for the shipping company, focusing on minimizing the total net cost. This net cost represents the difference between operating costs and operating revenue. Consequently, the net cost objective can be expressed as minimizing the following function:

$$g(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{d}}(\mathbf{v})) = - \sum_{i \in [I]} \sum_{j \in [I]} \sum_{t \in [T]} r_{i,j} \left(\tilde{d}_{i,j,t}(\mathbf{v}) \wedge \sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a^{\text{frgt}} L \right) + \sum_{a \in \mathcal{A}^{\text{else}}} c_a y_a. \quad (1)$$

Objective function (1) partitions the arcs into two categories. The first category consists of the freighting arcs, denoted by $\mathcal{A}^{\text{frgt}}$. For these arcs, the associated revenue is determined by the unit shipping revenue and the actual freight volume transported, i.e., $\sum_{i,j,t} r_{i,j} \tilde{d}_{i,j,t}(\mathbf{v}) \wedge \sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a^{\text{frgt}} L$. Here, $\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a^{\text{frgt}} L$ represents the total freight-carrying capacity assigned to the freighting arcs for origin-destination pair (i, j) in period t , where $\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a^{\text{frgt}}$ is the number of AESs assigned to these arcs and L is the capacity of each AES. If this total capacity is smaller than the realized freight demand $\tilde{d}_{i,j,t}(\mathbf{v})$, which depends on the contextual information $\mathbf{v}$, then part of the demand cannot be fulfilled; if it exceeds the realized demand, then some AES capacity remains underutilized. Therefore, the actual freight volume is given by $\tilde{d}_{i,j,t}(\mathbf{v}) \wedge \sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a^{\text{frgt}} L$. The second category includes the remaining four types of arcs, namely

Table 1
Types of arcs in $\mathcal{A}^{\text{else}}$ and their cost per unit flow.

Types of arc $a \in \mathcal{A}^{\text{else}}$	Cost per unit flow c_a
Relocation arc $(n_{i,j,b}, n_{j,i+l_{i,j},b-e^k_{i,j}})$	$c_{i,j}^{\text{relo}} l_{i,j}$
Idle arc $(n_{i,j,b}, n_{i,j+1,b})$	c_i^{idle}
Charging arc $(n_{i,j,b}, n_{i,j+1,b+e^c})$	$p_i^c + c^{\text{deg}}$
Discharging arc $(n_{i,j,b}, n_{i,j+1,b-e^d})$	$-(p_i^d - c^{\text{deg}})$

relocating arcs, charging arcs, discharging arcs, and idle arcs, which are collectively denoted by $\mathcal{A}^{\text{else}} := \mathcal{A}^{\text{relo}} \cup \mathcal{A}^{\text{idle}} \cup \mathcal{A}^{\text{chrg}} \cup \mathcal{A}^{\text{dsch}}$. The costs associated with these arcs are determined directly by their corresponding per-unit flow cost or revenue and the associated arc flows, and the unit flow costs of all four arc types in $\mathcal{A}^{\text{else}}$ are summarized in Table 1.

Operational constraints. We proceed to develop a dynamic AESs scheduling model that embeds the shipping company operator's operational constraints. The model explicitly accounts for the budget available for chartering AESs, the proper initialization of AESs deployment, and the flow balance of AESs throughout the spatial-temporal-SoC network.

1. Budget. In practice, the shipping company operates under limited financial resources when chartering AESs. To represent this operational reality, we impose a budget constraint that restricts the total chartering for AESs can be expressed as follows:

$$\sum_{i \in [I]} c_i x_i \leq M. \quad (2)$$

2. Initial period. At the beginning of the planning horizon, all AESs at the ports are assumed to be fully charged. Meanwhile, let $\delta^+(n_{i,j,b})$ and $\delta^-(n_{i,j,b})$ denote the sets of arcs for which $n_{i,j,b}$ is the origin and destination node respectively. To ensure consistency between the initial AESs availability and the movement decisions, the number of AESs moving from each port in the first period must equal the number of AESs initially assigned to that port. It is worth noting that, as shown in Section 3.2, AESs idling at the port also be regarded as a class of movements. This relationship is enforced through the following constraint:

$$\sum_{a \in \delta^+(n_{i,1,B})} (y_a^{\text{frgt}} + y_a^{\text{else}}) = x_i, \forall i \in [I]. \quad (3)$$

3. Flow balance. To maintain the consistency of AESs movements across the network $\mathcal{G}$, our model enforces flow balance at each node. For every intermediate time period, the number of AESs departing from a node must equal the number of AESs arriving at that node resulting in the following flow balance constraint:

$$\sum_{a \in \delta^+(n_{i,t,b})} (y_a^{\text{frgt}} + y_a^{\text{else}}) - \sum_{a \in \delta^-(n_{i,t,b})} (y_a^{\text{frgt}} + y_a^{\text{else}}) = 0, \forall i \in [I], t \in [T] \setminus \{1, T\}, b \in [B]. \quad (4)$$

4. Domain. To ensure feasibility and interpretability, we impose domain restrictions on the decision variables. In principle, the chartering variables x_i and arc-flow variables y_a should be restricted to nonnegative integers. To preserve computational tractability, however, we relax these integrality requirements and allow both x_i and y_a to take nonnegative continuous values. In practice, integer solutions can be approximately recovered from the resulting continuous solutions through a floor-rounding procedure (Nemhauser and Wolsey, 1988). Accordingly, we impose the following domain constraints:

$$y_a^{\text{frgt}} \geq 0, \forall a \in \mathcal{A}^{\text{frgt}}; y_a^{\text{else}} \geq 0, \forall a \in \mathcal{A}^{\text{else}} \quad (5)$$

$$x_i \geq 0, \forall i \in [I]. \quad (6)$$

In summary, we present a decision-making model for shipping company operators to optimize the dynamic AES scheduling problem, taking into account the uncertainty in freight demand. The model incorporates net cost objective while considering budget, initial period, and flow balance constraints. For conciseness, we present the dynamic AESs scheduling model as follows:

$$\begin{aligned} \min \quad & g(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{d}}(\mathbf{v})) \\ \text{s.t.} \quad & (2) - (6). \end{aligned} \quad (7)$$

However, the dynamic AESs scheduling model presents significant computational challenges for two main reasons: (i) it involves uncertain freight demand $\tilde{d}_{i,j,t}(\mathbf{v})$ that affects the objective function $g(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{d}}(\mathbf{v}))$, although correlated contextual information $\mathbf{v}$ can be exploited; (ii) the non-convex nature of the objective function $g(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{d}}(\mathbf{v}))$ complicates the reformulation. The first challenge is addressed by constructing a contextual JEPO framework, as detailed in Section 4. To tackle the second challenge, we propose a linear decision rule approximation that reformulates the non-convex constraint as a linear problem, as discussed in Section 5.

4. Contextual joint estimation-prediction-optimization framework

This section emphasizes the inherent uncertainty that exists on the freight demand. Based on the modeling configuration outlined in Section 3, the value of the uncertain demand $\tilde{\mathbf{d}}$ is not precisely known, as decisions are made before the realization of uncertainty. However, we have access to a collection of historical data that can be used to predict the demand. We denote the dataset as $\mathcal{K} = \{(\tilde{\mathbf{d}}_s, \hat{\mathbf{v}}_s)\}_{s \in [S]}$, which comprises S samples. The data is represented as $(\tilde{\mathbf{d}}_s, \hat{\mathbf{v}}_s) \in \mathbb{R}^{I^2 T} \times \mathbb{R}^K$ for $s \in [S]$, where $\tilde{\mathbf{d}}_s$ denotes the historical

freight demand and $\hat{v}_s$ denotes the contextual information associated with $\hat{d}_s$, including factors such as weather conditions and port throughput. Accordingly, the joint empirical distribution of demand and contextual information can be constructed as (Kannan et al., 2024):

$$\hat{\mathbb{P}}(\tilde{d}, \tilde{v}) = \frac{1}{S} \sum_{s \in [S]} \delta_{(\hat{d}_s, \hat{v}_s)},$$

where $\delta_{(\hat{d}_s, \hat{v}_s)}$ is the Dirac distribution concentrating unit mass at $(\tilde{d}, \tilde{v}) = (\hat{d}_s, \hat{v}_s)$.

A central challenge in robust optimization is how to choose an appropriate size of the uncertainty set. To address this issue, we adopt a target-oriented approach inspired by Brown and Sim (2009) and Long et al. (2023). Instead of specifying the uncertainty-set size a priori, we introduce a target value τ , which represents the maximum allowable net cost relative to the deterministic benchmark. We then seek scheduling decisions that remain feasible and attain a net cost no greater than τ over the largest possible uncertainty set. Under this formulation, the model maximizes the robustness of the scheduling decisions against demand uncertainty while ensuring that the total net cost remains within an acceptable level. At the decision-making stage, although the realized freight demand $\tilde{d} \in \mathcal{D} \subseteq \mathbb{R}^{I^2T}$ is unknown, the contextual information $\nu \in \mathbb{R}^K$, including factors such as weather conditions and port throughput, is observable. Accordingly, we propose the contextual joint estimation-prediction-optimization (JEPO) model of the dynamic AES scheduling problem as follows:

$$\begin{aligned} \Pi = \max_{\gamma \geq 0} \quad & \gamma \\ \text{s.t.} \quad & (2) - (6) \end{aligned} \tag{8a}$$

$$\begin{aligned} \sum_{a \in \mathcal{A}^{\text{else}}} c_a y_a - \sum_{i \in [I]} \sum_{j \in [I]} \sum_{t \in [T]} r_{i,j} \left(\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - \left[\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - \tilde{d}_{i,j,t}(\nu) \right]^+ \right) \leq \tau, \\ \forall \tilde{d}(\nu) \in \mathcal{D}(\gamma, \nu). \end{aligned} \tag{8b}$$

Constraint (8b) ensures that, for a given target value τ , the net cost does not exceed τ , where $[\cdot]^+ := \max\{\cdot, 0\}$. In particular, the term $\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - \left[\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - \tilde{d}_{i,j,t}(\nu) \right]^+$ represents the actual freight volume transported, and is equivalent to $\tilde{d}_{i,j,t}(\nu) \wedge \sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L$ in the net cost objective function (1), where $\wedge$ denotes the minimum operator.

In the above model, $\mathcal{D}(\gamma, \nu)$ is the JEPO uncertainty set proposed in this paper, which is defined in detail in Section 4.2, where γ denotes the deviation tolerance which governs the size of the uncertainty set and ν represents the contextual information at the moment of decision-making. The above model maximizes γ under the target τ , where τ is the objective value of the deterministic model.

The construction of the uncertainty set determines the trade-off between robustness and conservatism, and consequently affects the reliability and effectiveness of the solutions. The conventional uncertainty set is usually expressed as $\mathcal{U}(\gamma) = \{\tilde{d} \in \mathcal{D} \mid \|\tilde{d} - \hat{d}\|_p \leq \gamma\}$ using a classical norm-based metric $\|\cdot\|_p$, where $\hat{d}$ denotes the demand predicted from empirical data. The rest of this section is organized as follows, Section 4.1 introduces a novel contextual disparity metric to calculate the deviation between predicted demand and real demand. In Section 4.2, we construct the JEPO uncertainty set. Finally, in Section 4.3, we present a data-driven calibration of JEPO uncertainty set.

4.1. Contextual disparity metric

In this subsection, we focus on the construction of a contextual disparity metric, thereby allowing deviations to be evaluated in a more principled manner that reflects dimensional heterogeneity. In our problem, the value of freight demand $\tilde{d} \in \mathbb{R}^{I^2T}$ remains unknown, but the associated contextual information $\nu \in \mathbb{R}^K$ is observed. To incorporate this contextual information, we adopt an affine prediction map that links the contextual features to the predicted demand $\hat{d}^\dagger \in \mathbb{R}^{I^2T}$, given by $\hat{d}^\dagger := d_0 + \mathbf{D}\nu$, where $d_0 \in \mathbb{R}^{I^2T}$ and $\mathbf{D} \in \mathbb{R}^{I^2T \times K}$ are parameters estimated from the dataset.

Since the predicted demand may be deviated from the real demand $\tilde{d}$, we introduce a disparity metric $\rho_{\mathbf{H}}(\tilde{d}, \hat{d}^\dagger)$ which measures the deviation between $\tilde{d}$ and $\hat{d}^\dagger$. Conventionally, $\|\tilde{d} - \hat{d}^\dagger\|_p$ is employed to measure the deviation between $\tilde{d}$ and $\hat{d}^\dagger$, which solely depends on the fixed predicted output without considering contextual information. To better reflect the influence of contextual factors on the demand deviation, we introduce a contextual disparity metric, detailed in Definition 1.

Definition 1. (Contextual disparity metric) Given the L_p norm, the contextual disparity metric $\rho : \mathbb{R}^{I^2T} \times \mathbb{R}^{I^2T} \rightarrow [0, \infty]$ is defined as follows:

$$\rho_{\mathbf{H}}(\tilde{d}, \hat{d}^\dagger) := \|\mathbf{H}^{-1}(\tilde{d} - d_0 - \mathbf{D}\nu)\|_p,$$

where $\nu \in \mathbb{R}^K$ represents the contextual information and $\mathbf{H}$ represents the deviation matrix which is both symmetric and positive definite, i.e., $\mathbf{H} \in \mathbb{S}_{++}^{I^2T}$.

Remark 1. The deviation matrix $\mathbf{H}$ generalizes two special cases. When $\mathbf{H} = \mathbf{I}$, the contextual disparity metric reduces to the standard L_p norm $\|\tilde{d} - \hat{d}^\dagger\|_p$, treating all demand components equally regardless of their variability. This corresponds to the classical uncertainty set in robust optimization. When $\mathbf{H} = \mathbf{\Lambda}$, where $\mathbf{\Lambda}$ is a diagonal matrix with entries Λ_{ii} given by the standard deviation

of $\hat{d}_i^{(s)}$ across $s \in [S]$, the metric performs component-wise standardization, which is a common preprocessing technique in machine learning.

The deviation matrix $\mathbf{H}$ rescales the deviation vector before applying the L_p norm. When $\mathbf{H}$ is a diagonal matrix, $\mathbf{H}^{-1}\bar{\mathbf{d}}$ effectively normalizes the component-wise of $\bar{\mathbf{d}}$, a technique commonly used in machine learning. In our study, we allow $\mathbf{H}$ to be a dense matrix to adjust the weighting of deviations across ports and time periods, ensuring that infrequent but large variability in certain components is given relatively small weight, preventing it from dominating the overall deviation. Conversely, more stable components are assigned relatively higher weight. In this manner, the deviation matrix $\mathbf{H}$, estimated from the dataset, incorporates the heterogeneity of deviations into $\rho_{\mathbf{H}}(\cdot, \cdot)$, resulting in a more equitable and balanced measure of overall deviation. The following provides a specific example to illustrate the role of the deviation matrix $\mathbf{H}$.

Example 1. Consider two ports, denoted by A and B, over two time periods, denoted by t_1 and t_2 . The real demand is assumed to be $\bar{\mathbf{d}} = [d_{A,B,t_1}, d_{A,B,t_2}, d_{B,A,t_1}, d_{B,A,t_2}] = [100, 50, 70, 40]$, and the corresponding predicted demand vector is $\hat{\mathbf{d}}^\dagger = [d_{A,B,t_1}^\dagger, d_{A,B,t_2}^\dagger, d_{B,A,t_1}^\dagger, d_{B,A,t_2}^\dagger] = [90, 60, 65, 45]$. Accordingly, the demand deviation is $\bar{\mathbf{d}} - \hat{\mathbf{d}}^\dagger = [10, -10, 5, -5]$. Suppose that period t_1 is associated with high demand variability, with variance 4, whereas period t_2 is relatively stable, with variance 1. The resulting deviation matrix $\mathbf{H}$ obtained from the optimization model is given by

$$\mathbf{H} = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H}^{-1} = \begin{bmatrix} 0.25 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Here, $\mathbf{H}^{-1}$ assigns smaller penalties to deviations in high-variance periods and larger penalties to deviations in relatively stable periods. Accordingly, the weighted deviation is given by

$$\mathbf{H}^{-1}(\bar{\mathbf{d}} - \hat{\mathbf{d}}^\dagger) = \begin{bmatrix} 0.25 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ -10 \\ 5 \\ -5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -10 \\ 1.25 \\ -5 \end{bmatrix}.$$

Therefore, under the L_2 -norm, the contextual disparity metric is

$$\rho_{\mathbf{H}}(\bar{\mathbf{d}}, \hat{\mathbf{d}}^\dagger) = \sqrt{(2.5)^2 + (-10)^2 + (1.25)^2 + (-5)^2} = \sqrt{132.8125} \approx 11.5.$$

By contrast, without incorporating $\mathbf{H}$, the conventional Euclidean norm yields

$$\|\bar{\mathbf{d}} - \hat{\mathbf{d}}^\dagger\|_2 = \sqrt{10^2 + (-10)^2 + 5^2 + (-5)^2} = \sqrt{250} \approx 15.8.$$

This example shows that the matrix $\mathbf{H}$ adjusts the contribution of each component according to its variability. In particular, deviations in high-variance periods are downweighted, so that unusually volatile periods do not disproportionately dominate the overall disparity measure. As a result, the contextual disparity metric decreases from 15.8 to 11.5, yielding a more balanced and meaningful assessment of the deviation.

This example illustrates that $\mathbf{H}$ refines the deviation metric $\rho_{\mathbf{H}}(\cdot, \cdot)$ by incorporating weights, ensuring that the overall deviation is assessed in a fairer and more reasonable manner. Next, we describe the generalizability of the above definition deviation metric $\rho_{\mathbf{H}}(\cdot, \cdot)$.

Remark 2. The L_p -norm here can be extended to the absolute norm, and the application of the absolute norm is more extensive. An absolute norm satisfies $\|u\| = \|v\|$ for all $u \in \mathbb{R}^K$, with $v_\ell = |u_\ell|$ for each $\ell \in [L]$. The properties of the absolute norm are detailed in [Bertsimas and Sim \(2006\)](#). In particular, for any $v, u \in \mathbb{R}^K$ such that $v_\ell \geq |u_\ell|$ for all $\ell \in [L]$, $\|v\| \geq \|u\|$. Examples of absolute norms include the L_p -norm and the D -norm introduced in [Bertsimas and Sim \(2004\)](#).

In addition to the generalizability, we seek to show that the contextual disparity metric also has properties that are consistent with the common metric.

Theorem 1. The contextual disparity metric $\rho_{\mathbf{H}}(\cdot, \cdot)$ satisfies the following properties:

1. *Positive definite:* For all $\mathbf{d}, \boldsymbol{\zeta} \in \mathbb{R}^{I^2T}$, $\rho_{\mathbf{H}}(\mathbf{d}, \boldsymbol{\zeta}) \geq 0$. Moreover, $\rho_{\mathbf{H}}(\mathbf{d}, \boldsymbol{\zeta}) = 0$ if and only if $\mathbf{d} = \boldsymbol{\zeta}$.
2. *Convex:* $\rho_{\mathbf{H}}(\mathbf{d}, \boldsymbol{\zeta})$ is a convex function in $\mathbf{d}, \boldsymbol{\zeta} \in \mathcal{D}$.
3. *Translation invariant:* $\rho_{\mathbf{H}}(\mathbf{d}, \boldsymbol{\zeta}) = \rho(\mathbf{d} - \boldsymbol{\xi}, \boldsymbol{\zeta} - \boldsymbol{\xi})$ for all $\boldsymbol{\xi} \in \mathbb{R}^{I^2T}$.
4. *Positive homogeneous:* $\rho_{\mathbf{H}}(k\mathbf{d}, k\boldsymbol{\zeta}) = k\rho_{\mathbf{H}}(\mathbf{d}, \boldsymbol{\zeta})$ for all $k \geq 0$.
5. *Symmetric:* $\rho_{\mathbf{H}}(\mathbf{d}, \boldsymbol{\zeta}) = \rho_{\mathbf{H}}(\boldsymbol{\zeta}, \mathbf{d})$ for all $\mathbf{d}, \boldsymbol{\zeta} \in \mathbb{R}^{I^2T}$.

Proof. See E-Companion [Appendix B.1](#). $\square$

The first property indicates that the metric is always non-negative and only equals zero when the two vectors are identical which reflects the fundamental idea of a metric. The second property implies that the metric behaves well in optimization, ensuring that any local minimum is a global minimum, which simplifies solving optimization problems. The third property shows that shifting both vectors by the same amount does not affect the deviation metric which is important when data undergoes translation change. The fourth property ensures that if both vectors are scaled by the same factor, the deviation is scaled proportionally, which keeps the metric consistent across different magnitudes of data. The finally property indicates that the deviation metric is independent of the order of comparison, making the metric unbiased and consistent in its comparisons.

4.2. JEPO uncertainty set

Building on the contextual disparity metric discussed in [Section 4.1](#), we formally introduce the JEPO uncertainty set for the freight demand in this subsection. To ensure that the JEPO uncertainty set covers the uncertain demand $\tilde{d}$ with a high probability, it is essential to discuss the contextual confidence ball.

Definition 2. (Contextual confidence ball) Given deviation matrix $\mathbf{H}$, deviation tolerance γ , and parameters $\mathbf{d}_0$, $\mathbf{D}$, the contextual confidence ball of freight demand $\tilde{d}$ is defined as:

$$C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma) = \{\tilde{d} \in \mathcal{D} \mid \rho_{\mathbf{H}}(\tilde{d}, \hat{d}^*) \leq \gamma\} = \{\tilde{d} \in \mathcal{D} \mid \|\mathbf{H}^{-1}(\tilde{d} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\|_p \leq \gamma\},$$

centered at $(\mathbf{d}_0 + \mathbf{D}\mathbf{v})$. And the deviation tolerance γ controls the radius of the confidence ball.

To establish the upper bound of the infeasibility probability of the confidence, such that the uncertainty demand $\tilde{d}$ falls into the contextual confidence ball with a relatively high probability, we introduce the following theorem.

Theorem 2. (Probabilistic guarantee) Suppose the random variable $\tilde{d} \sim \mathbb{P}$, $\mathbb{P} \in \mathcal{P}_0(\mathcal{D})$, satisfies

$$\mathbb{E}_{\mathbb{P}}[\psi(\|\mathbf{H}^{-1}(\tilde{d} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\|_p)] \leq 1,$$

for some $\mathbf{d}_0 \in \mathbb{R}^{I^2T}$, $\mathbf{D} \in \mathbb{R}^{N \times K}$ and $\mathbf{H} \in \mathbb{S}_{++}^{I^2T}$, where $\psi(\cdot)$ is a power deviation penalty function which is defined in [Definition 3](#). Then we have the infeasibility probability bound of the confidence ball:

$$\mathbb{P}\{\tilde{d} \notin C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)\} \leq 1/\psi(\gamma) \quad \forall \gamma \geq 0.$$

Proof. See E-Companion [Appendix B.2](#). $\square$

We proceed by introducing the power deviation penalty function which serves as a key part in the above probabilistic guarantee.

Definition 3. (Power deviation penalty function) The power deviation penalty function $\psi : \mathbb{R}^+ \rightarrow [0, \infty]$ serves as a mapping relationship, mapping the deviation into a quantitative penalty value. This function is a non-negative, non-decreasing, and convex function such that $\psi(0) < 1$ and for some $\alpha \in [1, \infty]$, the power deviation penalty function is defined as follows:

$$\psi(\|\mathbf{H}^{-1}(\tilde{d} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\|_p) = \frac{\|\mathbf{H}^{-1}(\tilde{d} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\|_p^\alpha}{N},$$

which is normalized by N which represents the dimension of the uncertain parameter $\tilde{d}$.

In [Theorem 2](#), since the function $\psi(\cdot)$ is monotonically increasing, as γ approaches infinity, the probability of violating the feasibility bound becomes negligible. This suggests that the uncertain demand $\tilde{d}$ is highly likely to fall within the confidence ball.

Additionally, [Theorem 2](#) establishes a relationship between the infeasibility probability bound of the confidence ball $C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)$ and the estimated parameters $(\mathbf{d}_0, \mathbf{D}, \mathbf{H})$. Specifically, among the values of $(\mathbf{d}_0, \mathbf{D}, \mathbf{H})$ that satisfy the condition $\mathbb{E}_{\mathbb{P}}[\psi(\|\mathbf{H}^{-1}(\tilde{d} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\|_p)] \leq 1$, we aim to establish an upper bound on the volume of the confidence ball $C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)$, i.e., $\text{Vol}(C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma))$. If the volume of the confidence ball $C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)$ is too large, it means that more extreme uncertainty parameters need to be considered, resulting in overly conservative solutions obtained through optimization. For example, in resource allocation problems, excessive resources need to be reserved to deal with extreme situations, thereby reducing profits. The minimal volume of the confidence ball constrains the range of uncertainty vectors while ensuring uncertainty coverage, thereby reducing the cost of “fighting uncertainty”.

Building on the contextual disparity metric introduced in [Section 4.1](#) and the contextual confidence ball defined above, we define the JEPO uncertainty set as the minimum-volume confidence ball that preserves the prescribed confidence level, thereby mitigating the excessive conservatism of robust solutions.

Definition 4. (JEPO uncertainty set) Let $\mathbf{H} \in \mathbb{S}_{++}^{I^2T}$, $\mathbf{D} \in \mathbb{R}^{I^2T \times K}$, and $\mathbf{d}_0 \in \mathbb{R}^{I^2T}$ denote the parameters endogenously determined by the minimum volume confidence ball (MVCB) problem (10), which will be introduced below. We define the JEPO uncertainty set for the uncertain demand as

$$\mathcal{D}(\gamma, \mathbf{v}) = \left\{ \tilde{d} \in \mathcal{D} \mid \begin{array}{l} \rho_{\mathbf{H}}(\tilde{d}, \mathbf{d}_0 + \mathbf{D}\mathbf{v}) \leq \gamma \\ \mathbb{P}\{\tilde{d} \in C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)\} \geq 1 - \frac{1}{\psi(\gamma)} \\ \text{Vol}(C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)) \leq \hat{R} \end{array} \right\}, \quad (9)$$

where $\hat{R} = \text{Vol}(C(\mathbf{d}_0^*, \mathbf{D}^* \mathbf{v}, \mathbf{H}^*, \gamma))$, and $(\mathbf{d}_0^*, \mathbf{D}^*, \mathbf{H}^*)$ denotes an optimal solution to the MVCB problem (10).

In the JEPO uncertainty set $\mathcal{D}(\gamma, \mathbf{v})$, the first constraint requires that the contextual disparity metric $\rho_{\mathbf{H}}(\cdot, \cdot)$, as defined in [Definition 1](#), between the realized demand $\tilde{d}$ and the predicted demand $\mathbf{d}_0 + \mathbf{D}\mathbf{v}$ does not exceed the deviation tolerance γ . This constraint ensures that the uncertainty set excludes demand realizations that deviate excessively from the context-dependent prediction. The second constraint requires that the confidence ball $C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)$ covers the realized demand $\tilde{d}$ with probability at least $1 - 1/\psi(\gamma)$, where $\psi(\cdot)$ is the power deviation penalty function introduced in [Definition 3](#). Since $\psi(\gamma)$ increases monotonically with γ , the associated coverage probability approaches one as γ becomes larger, thereby ensuring the statistical reliability of the uncertainty set. The

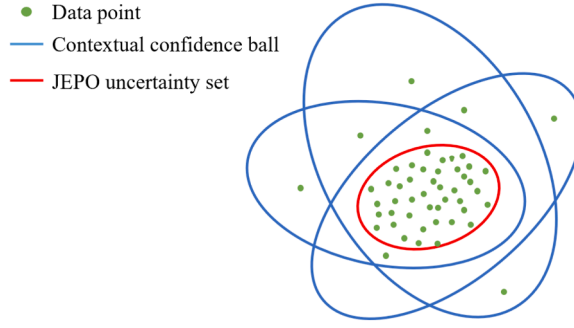
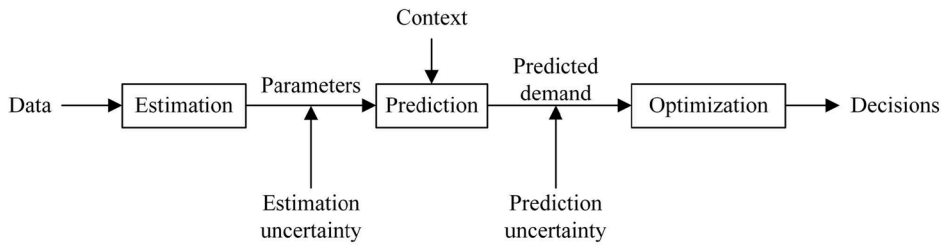
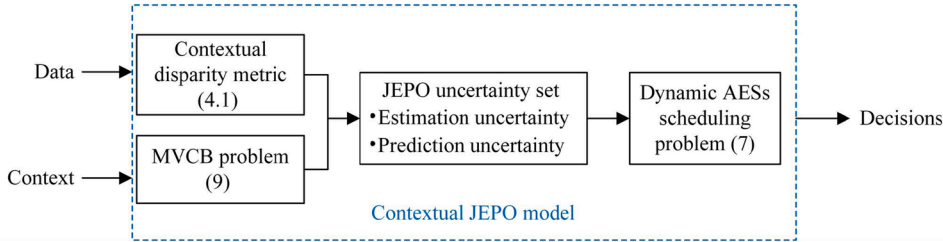


Fig. 2. JEPO uncertainty set and contextual confidence balls.

Note: Contextual confidence balls characterize the range in which uncertainty demand may vary for given parameters, whereas the JEPO uncertainty set is the particular confidence ball that both meets the required coverage probability and achieves the smallest volume.



(a) Sequential decision pipeline.



(b) Contextual JEPO framework.

Fig. 3. The process from the dataset to decisions.

Note: Compared with the sequential decision pipeline, the contextual JEPO framework embeds both estimation and prediction directly into the optimization model, forming an integrated process. Meanwhile, conventional parameter estimation typically minimizes a loss function, which may lead to mismatch with the final optimizing objective. Contextual JEPO framework estimates parameters by minimizing the volume of the confidence set, ensuring that the uncertainty set covers the uncertain demand with high probability while simultaneously reducing the conservatism of the solutions.

third constraint imposes an upper bound $\hat{R}$ on the volume of the confidence ball, which prevents the uncertainty set from becoming excessively large and overly conservative, and thus avoids unnecessarily costly robust decisions.

As illustrated in Fig. 2, we note that the JEPO uncertainty set differs from the contextual confidence balls. The contextual confidence balls describe the range where the uncertainty demand may fluctuate under given parameters $(d_0, \mathbf{D}, \mathbf{H})$, while the JEPO uncertainty set is the confidence ball that satisfies both probability and volume constraints among all such confidence balls, i.e., the confidence ball that provides sufficient coverage while being the most compact.

Under the JEPO uncertainty set, model (8) incorporates the parameters estimation, demand prediction, and decisions optimization in an unified framework. As illustrated in Fig. 3, the sequential decision-making process involves multiple independent stages, including estimation, prediction, and optimization. The interactions between these stages are often complex, which can result in high computational costs and the mismatch of the objective function. In contrast, the JEPO framework simplifies and integrates the decision-making process by directly feeding both contextual information and dataset into the model simultaneously. By focusing the entire process on a unified optimization objective, JEPO enhances the consistency and reliability of the resulting decisions.

However, managing the volume of the confidence ball $\text{Vol}(C(d_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma))$ poses a significant challenge. To overcome this, the following proposition offers an equivalent reformulation.

Proposition 1. The volume of $C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)$ can be precisely calculated as following by changing variables from $\mathbf{H}^{-1}(\tilde{\mathbf{d}} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})$ to $\mathbf{w}$:

$$\begin{aligned} \text{Vol}(C(\mathbf{d}_0 + \mathbf{D}\mathbf{v}, \mathbf{H}, \gamma)) &= \int_{\{\tilde{\mathbf{d}}: \|\mathbf{H}^{-1}(\tilde{\mathbf{d}} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\| \leq \gamma\}} d(\tilde{\mathbf{d}}) = \int_{\{\mathbf{w}: \|\mathbf{w}\| \leq \gamma\}} |\det(\mathbf{H})| d\mathbf{w} \\ &= |\det(\mathbf{H})| \cdot \text{Vol}(C(\mathbf{0}, \mathbf{I}, \gamma)) \propto |\det(\mathbf{H})|. \end{aligned}$$

This expression indicates that, for a fixed γ , the volume and therefore the size of each confidence ball is directly proportional to $\det(\mathbf{H})$. Consequently, selecting a deviation matrix with a smaller determinant yields a less conservative uncertainty set. Consistent with the probabilistic guarantee in [Theorem 2](#), the optimal parameter values can be obtained by solving the following optimization problem.

Proposition 2. The minimum volume confidence ball (MVCB) problem of the JEPO uncertainty set is:

$$\inf \quad \det(\mathbf{H}) \quad (10a)$$

$$\text{s.t.} \quad \mathbb{E}_{\mathbb{P}} [\psi(\|\mathbf{H}^{-1}(\tilde{\mathbf{d}} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\|_p)] \leq 1 \quad (10b)$$

$$\mathbf{H} \geq \text{Diag}(\mathbf{r}) \quad (10c)$$

$$\mathbf{d}_0 \in \mathbb{R}^{I^2 T}, \mathbf{D} \in \mathbb{R}^{N \times K}, \mathbf{H} \in \mathbb{S}_{++}^{I^2 T}, \quad (10d)$$

where the conic inequality $\mathbf{H} \geq \text{Diag}(\mathbf{r})$ represents $\mathbf{H} - \text{Diag}(\mathbf{r}) \in \mathbb{S}_{++}^{I^2 T}$ and the pre-specified regularization vector $\mathbf{r} \in \mathbb{R}_{++}^{I^2 T} \cup \{\mathbf{0}\}$ is applied to enforce $\mathbf{H} \in \mathbb{S}_{++}^{I^2 T}$, ensuring the volume of the confidence ball remains positive.

The JEPO uncertainty set $\mathcal{D}(\gamma, \mathbf{v})$ involves probability and volume constraints, which are challenging to handle directly in optimization. By introducing [Proposition 2](#) and replacing the implicit probability and volume bound constraint with explicit determinant and expectation constraints. Mathematically, the reformulated set $\tilde{\mathcal{D}}(\gamma, \mathbf{v})$ provides a more intuitive form, as shown in the following proposition.

Proposition 3. Given the $(\gamma, \mathbf{v})$, the JEPO uncertainty set (9) is equivalent to the following reformulation:

$$\tilde{\mathcal{D}}(\gamma, \mathbf{v}) = \left\{ \tilde{\mathbf{d}} \in \mathcal{D} \left| \begin{array}{l} \rho_{\mathbf{H}}(\tilde{\mathbf{d}}, \mathbf{d}_0 + \mathbf{D}\mathbf{v}) \leq \gamma \\ \mathbb{E}_{\mathbb{P}} [\psi(\|\mathbf{H}^{-1}(\tilde{\mathbf{d}} - \mathbf{d}_0 - \mathbf{D}\mathbf{v})\|_p)] \leq 1 \\ \det(\mathbf{H}) \leq \hat{K} \\ \mathbf{H} \geq \text{Diag}(\mathbf{r}) \\ \mathbf{d}_0 \in \mathbb{R}^{I^2 T}, \mathbf{D} \in \mathbb{R}^{N \times K}, \mathbf{H} \in \mathbb{S}_{++}^{I^2 T} \end{array} \right. \right\}, \quad (11)$$

where $\mathbf{r} \in \mathbb{R}_{++}^N \cup \{\mathbf{0}\}$ represents regularization vector. Let $\hat{K} = \det(\mathbf{H}^*)$, where $\mathbf{H}^*$ denotes the optimal solution set for the MVCB problem.

The uncertainty set (11) integrates parameter estimation and demand prediction within a unified framework. In the following subsection, we present a data-driven calibration of the JEPO uncertainty set.

4.3. Data-driven calibration of JEPO uncertainty set

In this subsection, we propose a data-driven calibration of JEPO uncertainty set, which provides a computationally tractable method for estimating the parameters $(\mathbf{d}_0, \mathbf{D}, \mathbf{H})$ of the JEPO uncertainty set (11). Since the true joint distribution is not available in practice, we use the empirical distribution $\hat{\mathbb{P}}(\tilde{\mathbf{d}}, \tilde{\mathbf{v}})$ to estimate these parameters. Furthermore, because problem (10) is non-convex and cannot be solved directly, we demonstrate how the MVCB problem can be reformulated as a convex optimization problem. This reformulation is presented in the following theorem.

Theorem 3. The optimal solution $(\mathbf{d}_0^*, \mathbf{D}^*, \mathbf{H}^*)$ of Problem (10) can be obtained from solving the following convex optimization problem:

$$\sup \quad \det(\mathbf{Q})^{1/N} \quad (12a)$$

$$\text{s.t.} \quad \mathbb{E}_{(\mathbf{d}_s, \mathbf{v}_s) \sim \hat{\mathbb{P}}(\tilde{\mathbf{d}}, \tilde{\mathbf{v}})} [\psi(\|\mathbf{Q}\mathbf{d}_s - \mathbf{q} - \mathbf{P}\mathbf{v}_s\|_p)] \leq 1 \quad (12b)$$

$$\mathbf{Q} \leq \text{Diag}^{-1}(\mathbf{r}) \quad (12c)$$

$$\mathbf{q} \in \mathbb{R}^{I^2 T}, \mathbf{P} \in \mathbb{R}^{N \times K}, \mathbf{Q} \in \mathbb{S}_{++}^{I^2 T}. \quad (12d)$$

Specifically, suppose the optimal solution of Problem (12) is $(\mathbf{q}^*, \mathbf{P}^*, \mathbf{Q}^*)$. Then $\mathbf{H}^* = (\mathbf{Q}^*)^{-1}$, $\mathbf{d}_0^* = \mathbf{H}^* \mathbf{q}^*$, and $\mathbf{D}^* = \mathbf{H}^* \mathbf{P}^*$.

Proof. See E-Companion [Appendix B.3](#). $\square$

It is well known that the determinant optimization problem in the above issue can be expressed as the semidefinite optimization problem with a geometric mean constraint which can subsequently expressed as second-order cone problem (SOCP). These methods are implemented in RSOME, a Python-based algebraic modeling toolbox developed by [Chen and Xiong \(2023\)](#). Beyond its convexity, problem (12) also benefits from the tractability of computing the expectation over the empirical distribution in the constraint, as following:

$$\mathbb{E}_{(\mathbf{d}_s, \mathbf{v}_s) \sim \hat{\mathbb{P}}(\tilde{\mathbf{d}}, \tilde{\mathbf{v}})} [\psi(\|\mathbf{Q}\mathbf{d}_s - \mathbf{q} - \mathbf{P}\mathbf{v}_s\|_p)] = \frac{1}{S} \sum_{s \in [S]} \psi(\|\mathbf{Q}\mathbf{d}_s - \mathbf{q} - \mathbf{P}\mathbf{v}_s\|_p).$$

We next focus on a special case, we set the norm to be the L_2 -norm and the power deviation penalty function to be $\psi_2(s)$. Therefore, we derive the following theorem.

Theorem 4. *The optimal solution to model (12) for the specific case with an L_2 -norm, a deviation penalty $\psi_2(\cdot)$, and $\mathbf{r} = \mathbf{0}$ can be expressed as follows:*

$$\mathbf{D}^* = \hat{\Sigma}_{dv} \hat{\Sigma}_v^{-1}, d_0^* = \hat{\mu}_d - \mathbf{D}^* \hat{\mu}_v, \text{ and } \mathbf{H}^* = \left(\hat{\Sigma}_d - \hat{\Sigma}_{dv} \hat{\Sigma}_v^{-1} \hat{\Sigma}_{vd} \right)^{\frac{1}{2}},$$

where $\hat{\mu}_v = \mathbb{E}_{\hat{\mathbb{P}}}[\tilde{v}]$, $\hat{\mu}_d = \mathbb{E}_{\hat{\mathbb{P}}}[\tilde{d}]$, $\hat{\Sigma}_v = \mathbb{E}_{\hat{\mathbb{P}}}[(\tilde{v} - \hat{\mu}_v)(\tilde{v} - \hat{\mu}_v)^\top]$, $\hat{\Sigma}_d = \mathbb{E}_{\hat{\mathbb{P}}}[(\tilde{d} - \hat{\mu}_d)(\tilde{d} - \hat{\mu}_d)^\top]$, and $\hat{\Sigma}_d - \hat{\Sigma}_{dv} \hat{\Sigma}_v^{-1} \hat{\Sigma}_{vd}$ is assumed to be positive definite.

Proof. See E-Companion Appendix B.4. $\square$

Through this theorem, we find that the optimal solution is completely determined by the first-order moment (mean) and second-order moment (covariance) under the empirical distribution $\hat{\mathbb{P}}$. Theorem 4 provides an optimal analytical solution that can be calculated directly without optimization under specific conditions.

Building on Theorem 3, we introduce a data-driven calibration of the uncertainty set in the following proposition. For convenience, we define $\Xi(d_s, v_s) := \{\det(\mathbf{Q})^{1/N} \mid (12b)-(12d)\}$.

Proposition 4. *Based on the solution to the optimization problem (12), the uncertainty set (11) can be reduced to the following concise reformulation:*

$$\hat{D}(\gamma, v) = \left\{ \tilde{d} \in D \mid \begin{array}{l} \|\mathbf{Q}^* \tilde{d} - q^* - \mathbf{P}^* v\|_p \leq \gamma \\ (q^*, \mathbf{P}^*, \mathbf{Q}^*) = \operatorname{argmax} \Xi(d_s, v_s) \end{array} \right\}, \quad (13)$$

where $(q^*, \mathbf{P}^*, \mathbf{Q}^*)$ is the optimal solution obtained by solving problem (12).

Proof. See E-Companion Appendix B.7. $\square$

As shown in Proposition 4, this calibration enables efficient estimation of the parameters by leveraging state-of-the-art solvers such as Rsome and Gurobi. Subsequent solutions to the dynamic AESs scheduling problem are derived using the data-driven calibration JEPO uncertainty set (13).

5. Tractability of contextual JEPO framework

In this section, we discuss how one can solve the JEPO model (8) with uncertainty set $\hat{D}(\gamma, v)$ in an effective way. Section 5.1 reformulates the nonlinear part of constraint (8b) into a linear solvable form using the LDR approximation. Next, Section 5.2 constructs a lifted uncertainty set to address the inconsistency existing of uncertain demand in constraint and uncertainty set. Finally, to handle the decision-dependent uncertainty, we propose a binary search algorithm for the deviation tolerance γ in Section 5.3.

5.1. Linear decision rule approximation

In this subsection, we concentrate on dealing with the nonlinear term of the constraint (8b). In the following, we apply the idea of LDR to seek a scalable approach to circumvent the computational difficulty. The LDR approximation is commonly used to tackle the intractability in robust optimization by limiting intractable term to affine functions of the random variables (He et al., 2020; Hao et al., 2020). In our problem, we apply this method to approximate the piecewise function $\left[\sum_{a \in D_{i,j,t}^{\text{frgt}}} y_a L - \tilde{d}_{i,j,t}(v) \right]^+$. Specifically, for all $i, j \in [I], t \in [T]$, we replace the piecewise function by $k_{i,j,t}(\tilde{d}_{i,j,[t-1]})$ and impose additional constraints $k_{i,j,t}(\tilde{d}_{i,j,[t-1]}) \geq \left[\sum_{a \in D_{i,j,t}^{\text{frgt}}} y_a L - \tilde{d}_{i,j,t}(v) \right]^+$, where

$$k_{i,j,t}(\tilde{d}_{i,j,[t-1]}) = k_{0,i,j,t} + \sum_{i' \in [I]} \sum_{j' \in [I]} \sum_{t'=1}^{t-1} (k_{1,i,j,t,i',j',t'} \tilde{d}_{i',j',t'}), \quad (14)$$

with the coefficients $k_{0,i,j,t}$ and $k_{1,i,j,t,i',j',t'}$ are the new decision variables. More compactly, the LDR approximation is represented as $k(\cdot) \in \mathcal{L}$ where

$$\mathcal{L} = \left\{ k(\cdot) : D \rightarrow \mathbb{R}^+ \mid \begin{array}{l} \text{for all } i, j \in [I], t \in [T] : \\ k_{i,j,t}(\tilde{d}_{i,j,[t-1]}) = k_{0,i,j,t} + \sum_{i' \in [I]} \sum_{j' \in [I]} \sum_{t'=1}^{t-1} (k_{1,i,j,t,i',j',t'} \tilde{d}_{i',j',t'}) \\ \text{for some } k_{0,i,j,t}, k_{1,i,j,t,i',j',t'} \in \mathbb{R}, i', j' \in [I], t' \in [1, t-1] \end{array} \right\}. \quad (15)$$

Note that $k_{i,j,t}(\tilde{d}_{i,j,[t-1]})$ depends on the demand that occurred during the first $t-1$ periods because the amount of freight demand needs to be completed in time period t is related to the demand in the first $t-1$ time periods. Utilizing the LDR approximation, we have the following affine approximation robust counterpart for the JEPO model in the following proposition.

Proposition 5. Given the JEPO uncertainty set $\hat{D}(\gamma, \mathbf{v})$ defined in (11) and the LDR approximation defined in (15), the JEPO model (8) can be equivalently reformulated as follows:

$$\hat{\Pi} = \max_{\gamma \geq 0} \gamma \quad (16a)$$

$$\text{s.t.} \quad (2) - (6)$$

$$\sum_{a \in \mathcal{A}^{\text{else}}} c_a y_a - \sum_{i \in [I]} \sum_{j \in [I]} \sum_{t \in [T]} r_{i,j} \left(\sum_{a \in \mathcal{D}_{i,j,t}^{\text{fgrt}}} y_a L - k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,t}[t-1]) \right) \leq \tau, \forall \tilde{\mathbf{d}} \in \hat{D}(\gamma, \mathbf{v}) \quad (16b)$$

$$k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,t}[t-1]) \geq \sum_{a \in \mathcal{D}_{i,j,t}^{\text{fgrt}}} y_a L - \tilde{\mathbf{d}}_{i,j,t}(\mathbf{v}), \forall i, j \in [I], t \in [T], \tilde{\mathbf{d}} \in \hat{D}(\gamma, \mathbf{v}) \quad (16c)$$

$$k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,t}[t-1]) \geq 0, \forall i, j \in [I], t \in [T], \tilde{\mathbf{d}} \in \hat{D}(\gamma, \mathbf{v}) \quad (16d)$$

$$\mathbf{k}(\cdot) \in \mathcal{L}. \quad (16e)$$

At the expense of introducing $O(N^2)$ decision variables, we linearize the sums of piecewise function terms.

Proposition 5 establishes an equivalent reformulation of the JEPO model under the LDR approximation. We next present a result to clarify the feasibility of this approximation.

Proposition 6. For any feasible decisions $\mathbf{x}$ and $\mathbf{y}$, and for any $\tilde{\mathbf{d}} \in \hat{D}(\gamma, \mathbf{v})$, there exists a decision rule $\mathbf{k}(\cdot) \in \mathcal{L}$ satisfying constraints (16b)–(16d); that is, the LDR approximation problem (16) admits a feasible solution.

Proof. See E-Companion [Appendix B.5](#). $\square$

While LDR resolves the nonlinear terms associated with demand, the mismatch between scalar-level constraints and the vector-level uncertainty set which leads to both theoretical complications and computational intractability. To reconcile the structure of the uncertain parameters in the constraints and in the uncertainty set, we introduce a lifted uncertainty set in the next subsection.

5.2. Lifted JEPO uncertainty set

In this subsection, we investigate a fundamental theoretical challenge stemming from the indivisible structure of the uncertainty set. In model (16), the function $k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,t}[t-1])$ appearing in constraints (16b)–(16d) depends linearly on the scalar component-wise $\tilde{\mathbf{d}}_{i,j,t}$ of the demand vector $\tilde{\mathbf{d}}$. However, the uncertainty set $\hat{D}(\gamma, \mathbf{v})$ characterizes $\tilde{\mathbf{d}}$ as an inseparable vector that cannot be decomposed across indices (i, j, t) . As a result, direct reformulation of the problem is not tractable for standard optimization solvers.

To address this inconsistency, we construct a lifted uncertainty set following the approach of [Chen and Xiong \(2023\)](#). This set expands the original uncertainty space into a higher-dimensional representation by introducing auxiliary random variables $\tilde{\mathbf{w}}_1$, $\tilde{\mathbf{w}}_2$, and $\tilde{u}$, defined as follows:

$$\mathcal{D}'(\gamma, \mathbf{v}) = \left\{ (\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \in \mathcal{D} \times \mathbb{R}_+^{I^2 T} \times \mathbb{R}_+^{I^2 T} \times \mathbb{R}_+ \left| \begin{array}{l} \tilde{\mathbf{w}}_1 - \tilde{\mathbf{w}}_2 = \mathbf{Q}^* \tilde{\mathbf{d}} - \mathbf{q}^* - \mathbf{P}^* \mathbf{v} \\ \|\tilde{\mathbf{w}}_1 + \tilde{\mathbf{w}}_2\|_p \leq \tilde{u} \leq \gamma \\ (\mathbf{q}^*, \mathbf{P}^*, \mathbf{Q}^*) = \arg \max \Xi(\mathbf{d}_s, \mathbf{v}_s) \end{array} \right. \right\}.$$

Proposition 7. The lifted uncertainty set $\mathcal{D}'(\gamma, \mathbf{v})$ with auxiliary random variables $\tilde{\mathbf{w}}_1$, $\tilde{\mathbf{w}}_2$ and $\tilde{u}$ is equivalent to the original set $\hat{D}(\gamma, \mathbf{v})$.

Proof. See E-Companion [Appendix B.8](#). $\square$

Based on the [Proposition 7](#) and standard duality argument, one can reformulate problem (16) with $\hat{D}(\gamma, \mathbf{v})$ into the following semi-infinite program in [Proposition \(8\)](#), which can then be reformulated as a second-order cone program (SOCP).

Proposition 8. The problem (16) with uncertainty set $\hat{D}(\gamma, \mathbf{v})$ is equivalent to the following optimization problem with the lifted uncertainty set $\mathcal{D}'(\gamma, \mathbf{v})$:

$$m(\tau) = \max_{\gamma \geq 0} \gamma \quad (17a)$$

$$\text{s.t.} \quad (2) - (6)$$

$$\sum_{a \in \mathcal{A}^{\text{else}}} c_a y_a - \sum_{i \in [I]} \sum_{j \in [I]} \sum_{t \in [T]} r_{i,j} \left(\sum_{a \in \mathcal{D}_{i,j,t}^{\text{fgrt}}} y_a L - k_{i,j,t}(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \right) \leq \tau, \quad (17b)$$

$$k_{i,j,t}(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \geq \sum_{a \in \mathcal{D}_{i,j,t}^{\text{fgrt}}} y_a L - \tilde{\mathbf{d}}_{i,j,t}(\mathbf{v}), \quad (17c)$$

$$\forall i, j \in [I], t \in [T], (\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \in \mathcal{D}'(\gamma, \mathbf{v}) \quad (17c)$$

$$k_{i,j,t}(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \geq 0, \forall i, j \in [I], t \in [T], (\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \in \mathcal{D}'(\gamma, \mathbf{v}) \quad (17d)$$

$$k(\cdot) \in \mathcal{A}(2N + 1). \quad (17e)$$

Here, $\mathcal{A}(2N + 1)$ denotes the set of all affine functions on domain $\mathbb{R}^{2N+1}$ to $\mathbb{R}$. That is to say, the LDR approximation (15) as an affine function of $\tilde{d}_{i,j,t}$, $i, j \in [I]$, $t \in [T]$ transforms into function $k_{i,j,t} : \mathbb{R}^{2N+1} \rightarrow \mathbb{R}$, as an affine function of vector $(\tilde{w}_1, \tilde{w}_2, \tilde{u})$, which is given by:

$$k_{i,j,t}(\tilde{w}_1, \tilde{w}_2, \tilde{u}) = k_{i,j,t,0} + \mathbf{k}_{i,j,t,1}^\top \tilde{w}_1 + \mathbf{k}_{i,j,t,2}^\top \tilde{w}_2 + k_{i,j,t,3} \tilde{u},$$

where $k_{i,j,t,0}, k_{i,j,t,3} \in \mathbb{R}$ and $\mathbf{k}_{i,j,t,1}, \mathbf{k}_{i,j,t,2} \in \mathbb{R}^{I^2 T}$, $[\mathbf{k}_{i,j,t,1}]_{i',j',t'} = [\mathbf{k}_{i,j,t,2}]_{i',j',t'} = 0, \forall i' \geq t$ which are affine functions with respect to lifted variables.

Proof. See E-Companion Appendix B.9. $\square$

Under the lifted uncertainty set, the reformulation makes the structure of the uncertain parameters consistent in the constraints and the uncertainty set, thereby establishing computational tractability under LDR. Nevertheless, a key challenge remains in maximizing the parameter γ of the uncertainty set, which lead to decision-dependent uncertainty. Since standard optimization solvers cannot directly handle this issue, in the next subsection we propose an binary search with adaptive interval algorithm to overcome this challenge.

5.3. Binary search with adaptive interval for deviation tolerance

To address the computational intractability caused by the decision-dependent structure of the uncertainty set, in this subsection we develop an binary search with adaptive interval algorithm. Since standard convex optimization solvers cannot directly handle uncertainty set that depend on decision variables, we reformulate the original problem as a sequence of parameterized subproblems. In each subproblem, the value of $\gamma \geq 0$ is fixed, and the problem is represented as follows:

$$m^\gamma(\tau) = \min_{\zeta \leq 0} \quad \zeta - \tau \quad (18a)$$

$$\text{s.t.} \quad (2) - (6); (17c) - (17e)$$

$$\sum_{a \in \mathcal{A}^{\text{else}}} c_a y_a - \sum_{i \in [I]} \sum_{j \in [I]} \sum_{t \in [T]} r_{ij} \left(\sum_{a \in D_{i,j,t}^{\text{rgt}}} y_a L - k_{i,j,t}(\tilde{w}_1, \tilde{w}_2, \tilde{u}) \right) \leq \zeta, \quad (18b)$$

$$\forall (\tilde{d}, \tilde{w}_1, \tilde{w}_2, \tilde{u}) \in D'(\gamma, v).$$

We exploit the monotonicity of the function $m^\gamma(\tau)$ with respect to γ to design a precision search method which converges to the largest feasible γ within a prescribed error.

Algorithm 1 Binary search with adaptive interval for deviation tolerance.

Require: Let $\bar{\gamma}$ be a sufficiently large initial upper bound such that $m^{\bar{\gamma}}(\tau) \geq 0$. If this condition is not satisfied, we iteratively expand $\bar{\gamma}$ until feasibility is guaranteed. Let $\Delta > 0$ denote the desired convergence accuracy.

1: **Initialization:** Set $\gamma_1 = 0$ and $\gamma_2 = \bar{\gamma}$.

2: **Iterative refinement:** While $\gamma_2 - \gamma_1 > \Delta$, update:

1. Set $\gamma := \frac{\gamma_1 + \gamma_2}{2}$;

2. Solve model (18) with fixed γ and compute $m^\gamma(\tau)$;

3. If $m^\gamma(\tau) \leq 0$, update $\gamma_1 := \gamma$; otherwise, set $\gamma_2 := \gamma$.

3: **Termination:** Output the solution $\mathbf{x}$ and y_a .

Proposition 9. Suppose that model (17) is feasible. Then, Algorithm 1 finds solutions $\mathbf{x}$ and y_a with objective $\tilde{\gamma}$ satisfying $|\tilde{\gamma} - \gamma^*| < \Delta$ in at most $\lceil \log_2(\bar{\gamma}/\Delta) \rceil$ computations of subproblem (18). Here, γ^* is the optimal objective of model (17), $\Delta > 0$ is an arbitrary number, and $\bar{\gamma} > 0$ is a sufficiently large number.

Proof. See E-Companion Appendix B.6. $\square$

Proposition 9 provides the theoretical guarantees for the proposed algorithm by providing an asymptotic consistency. The convergence condition $|\tilde{\gamma} - \gamma^*| < \Delta$ ensures that our algorithm preserves the solution quality of the optimization problem, demonstrating that the decomposition strategy does not compromise optimality within the specified tolerance. Additionally, the iteration bound $\lceil \log_2(\bar{\gamma}/\Delta) \rceil$ not only allows for accurate prediction of computational resource requirements but also highlights the logarithmic complexity advantage of the proposed algorithm. This bounded iteration guarantees the algorithm's termination in finite time, ensuring the computational reliability for practical.

6. Numerical studies

In this section, we outline the application of our methodology and investigate the performance of the model. We use the synthetic data to generate diverse instances with various freight demand, spatial-temporal-SoC network, electricity charging and discharging prices. Section 6.1 provides the detailed data description and parameter settings. In Section 6.2, we validate the out-of-sample

performance of the contextual JEPO framework. Section 6.3 draws insights on the benefit of integrating S2G into the dynamic AESs scheduling problem. Lastly, Section 6.4 conducts a case study of eight ports in the North Sea region, Europe and provides management insights.

6.1. Data description and parameter settings

Our research focuses on the North Sea region, Europe, one of Europe's most active maritime corridors, which hosts several major hub ports such as Rotterdam, Antwerp, and Hamburg. The region is characterized by dense port networks and a high level of maritime electrification, making it an ideal context for investigating the scheduling problem of AESs.

In our numerical experiments, we consider three ports Rotterdam, Antwerp, and Vlissingen in the North Sea region, Europe, i.e., $I = 3$. We set each time period cycle to one day, i.e., 24 hours, and consider five time periods, i.e., $T = 5$. And the battery capacity is 5 MWh, which is a typical AES battery capacity. We divide it into five SoC levels, i.e., $B = 5$. We consider both charging and discharging of rate 1 SoC unit/period, i.e., 1 MWh. We also consider the volatility of electricity prices across different time periods, with time-of-use charging and discharging prices assume to be $p_t^c = \{40, 65, 100, 80, 60\}$ and $p_t^s = \{80, 100, 150, 130, 105\}$, respectively. We also incorporate the battery degradation cost per unit of state of charge (SoC), denoted by c^{deg} , to internalize the long-term cost of battery wear within the optimization framework. In this study, we set $c^{\text{deg}} = 8\$/\text{MWh}$ based on the calculations reported in Peterson et al. (2010).

Unfortunately, the dataset lacks detailed information on historical freight demand, making it impractical to perform such a case study. Therefore, we use the available demand data and synthetic data to generate the freight demand d . To investigate the performance of our JEPO model under different charter budgets, we consider $M \in \{10000, 15000, 20000, 25000, 30000\}$ for the budget constraint (2). We prepare two demand datasets to examine the performance of the model: the (in-sample) training data to derive the decisions x_i, y_a , and the (out-of-sample) test data which is used to evaluate the out-of-sample performance of the decisions x_i, y_a .

Contextual information. As presented in Section 3.1, two main covariates we considered, which significantly influence freight demand, are weather fluctuations and port throughput. We gather the weather information of the ports in North Sea region, Europe from Weather Underground (WU, 2025). We also collect port information from the European Seaports Organization (ESPO, 2025). Therefore, we set the contextual information v as a 2-dimensional vector of weather and port throughput.

Target parameter. Following the target-oriented robust satisficing principle, we set this target using a deterministic contextual benchmark that reflects the reference cost the decision maker would expect when relying on the contextual prediction of freight demand. Specifically, for each budget level M and observed contextual vector v , we first solve a deterministic scheduling model in which the uncertain freight demand is replaced by its predicted value $f(v)$. Let $Z_0(M, v)$ denote the resulting optimal net cost, i.e.,

$$Z_0(M, v) = \min_{x, y} g(x, y, f(v)) \quad \text{s.t. (2) - (6)},$$

and we set $\tau = Z_0(M, v)$ accordingly.

Radius of the uncertainty set. Given the target parameter τ , the radius of the uncertainty set is represented by the deviation tolerance γ . Rather than fixing γ exogenously as a tuning parameter, we determine it endogenously as the largest deviation tolerance for which the target performance level τ can still be guaranteed. Specifically, we follow the binary search with adaptive interval procedure described in Algorithm 1. For a fixed candidate value of γ , we solve the corresponding parameterized robust subproblem and check whether the target constraint can be satisfied. If the subproblem is feasible, the lower bound of the search interval is updated to this candidate value; otherwise, the upper bound is updated and the search terminates when the interval length is smaller than the desired convergence accuracy.

All numerical experiments are conducted on a PC with a 3.6 GHz processor and 16 GB of RAM. The algorithms are coded in Python 3.13, and the optimization models were solved by the commercial solver Gurobi 12.0.1.

6.2. Out-of-sample performance of models

In this subsection, we evaluate the out-of-sample performance of the contextual JEPO model by comparing it with several benchmark models. To ensure a fair comparison, the decisions of all benchmark models are also made conditional on the available contextual information. Specifically, we estimate the mapping $f: \mathbb{R}^K \rightarrow \mathbb{R}^{I^2 T}$, which relates contextual information to freight demand, using ordinary least squares (OLS) on the dataset. The benchmark models are described as follows.

(i) The deterministic (DET) model, which ignores demand uncertainty and treats the predicted demand $f(v)$ as the realized demand.

(ii) The residual-based sample average approximation (SAA) model, in which the term $g(x, y, \tilde{d}(v))$ in model (7) is replaced by $\frac{1}{S} \sum_{s=1}^S g(x, y, f(v) + \hat{\varepsilon}_s)$, where $\hat{\varepsilon}_s := \hat{d}_s - f(\hat{v}_s)$, $s \in [S]$, denotes the empirical residual corresponding to sample s .

(iii) The contextual robust optimization (CRO) model, which adopts the standard uncertainty set $\mathcal{U}(\gamma) = \{\tilde{d} \in \mathcal{D} \mid \|\tilde{d} - \hat{d}\| \leq \gamma\}$, typically defined as a norm ball, where $\hat{d} = f(v)$ denotes the predicted demand under the current contextual information.

(iv) The standardized robust optimization (SRO) model, which fixes $\mathbf{H} = \mathbf{A}$, where each diagonal element Λ_{ii} is computed from the dataset as the sample standard deviation of the i -th demand component.

Utilizing the parameter configurations outlined in Section 6.1, we solve the JEPO model and benchmark models, yielding the corresponding optimal strategy (x_i, y_a) . Subsequently, we assess the out-of-sample performance using the Monte Carlo method with an additional 1000 test samples.

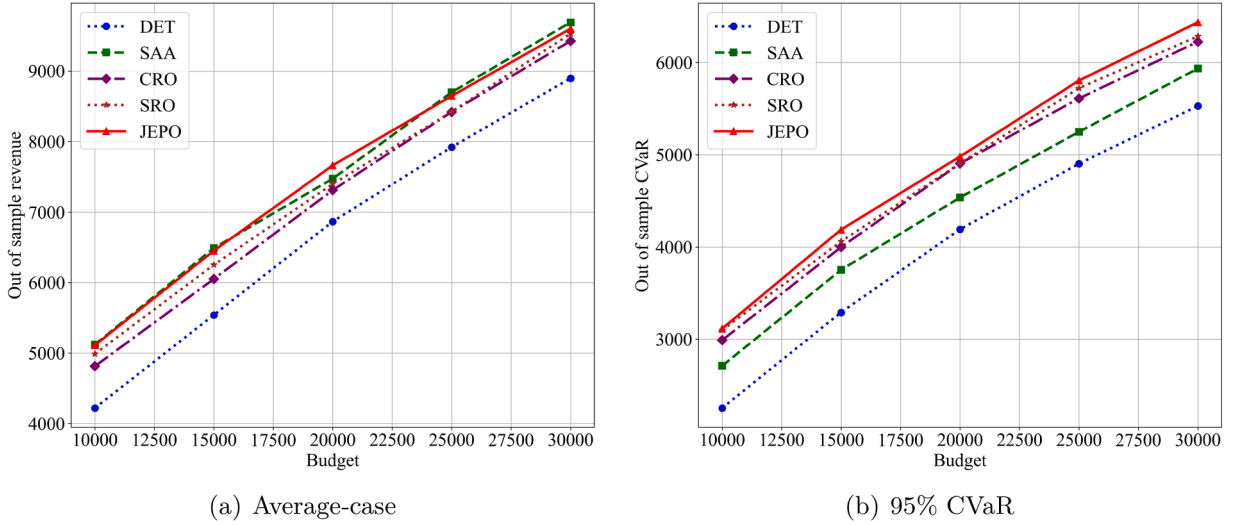


Fig. 4. Out-of-sample performance of our JEPO model and benchmark models under different budget.

We evaluate the out-of-sample performance of the profitability through two metrics, encompassing the average-case, 95% Conditional Value at Risk (CVaR). The average-case metric quantifies the average performance of models, while the 95% CVaR metric offers insights into their worst-case performance, i.e., the average of the 5% tail distribution.

For different budget levels $M \in \{10000, 15000, 20000, 25000, 30000\}$, Fig. 4 presents the out-of-sample performance of the proposed JEPO model and the three benchmark models with respect to the revenue objective under different evaluation metrics. It can be seen that the JEPO model consistently outperforms all benchmark models across the tested budget settings. This superior performance stems from its ability to address the key limitations of the benchmark approaches. Specifically, the deterministic model treats the predicted demand as the realized demand and therefore ignores demand uncertainty, which leads to poor performance when the actual demand deviates from the prediction. The SAA model approximates the realized demand by combining the prediction with empirical residual samples, but this approximation does not adequately capture the structure of contextual uncertainty, resulting in inferior out-of-sample performance. The CRO model relies on a standard box-type uncertainty set and does not explicitly account for the confidence level or control the conservatism of the uncertainty set, which also limits its effectiveness.

The SRO model differs from CRO by using a diagonal deviation matrix $\mathbf{H} = \mathbf{\Lambda}$ to perform component-wise standardization of demand deviations, and indeed outperforms CRO in most budget settings. However, the diagonal structure of $\mathbf{\Lambda}$ captures only the standard deviation of each demand component separately, while ignoring the cross-dimensional correlations among ports and time periods. By contrast, the JEPO model constructs the uncertainty set by explicitly incorporating both estimation and prediction procedures, and estimates a full matrix $\mathbf{H}^*$ via the MCVB approach to jointly capture demand heterogeneity and cross-dimensional correlations. This effectively reduces the excessive conservatism of robust decisions and thereby improves out-of-sample performance.

To further assess the robustness of the target specification, we conduct a sensitivity analysis by varying τ around the reference level $Z_0(M, v)$. Specifically, we consider $\tau = 0.85Z_0(M, v)$ and $\tau = 1.15Z_0(M, v)$ and evaluate the corresponding 95% CVaR out-of-sample performance across different budget levels. As shown in Fig. C.3, JEPO exhibits consistent out-of-sample performance under both target settings.

To understand why the JEPO model outperforms other models, Fig. 5 presents the initial AESs allocation across 8 ports when $M = 10000$. Compared with the benchmark models, the JEPO model tends to derive a mild allocation plan, i.e., it does not allocate too many AESs in some specific ports and thereby results in a stable revenue.

We also report the computational times of the JEPO model and the three benchmark models under different budget levels M in Table 2. As M increases from 10,000 to 30,000, the DET model remains the fastest, with a runtime of less than one second, while the SAA and CRO models require moderate and relatively stable solution times. Owing to the iterative nature of the bisection search procedure and the need to solve the MCVB estimation problem, the JEPO model requires a comparatively longer computational time. Nevertheless, its runtime increases only moderately, from 1764 to 2032 seconds, as the budget level increases. This indicates that the JEPO model maintains stable computational performance across different budget settings. Moreover, since the AES scheduling problem considered here is a strategic planning problem rather than a real-time operational problem, the observed computational time remains acceptable for practical implementation.

6.3. Benefit of S2G integration

In this subsection, we investigate various benefits of integrating S2G in the dynamic AESs scheduling problem as well as how the benefits are impacted by factors such as various budgets, seasonality of electricity prices.

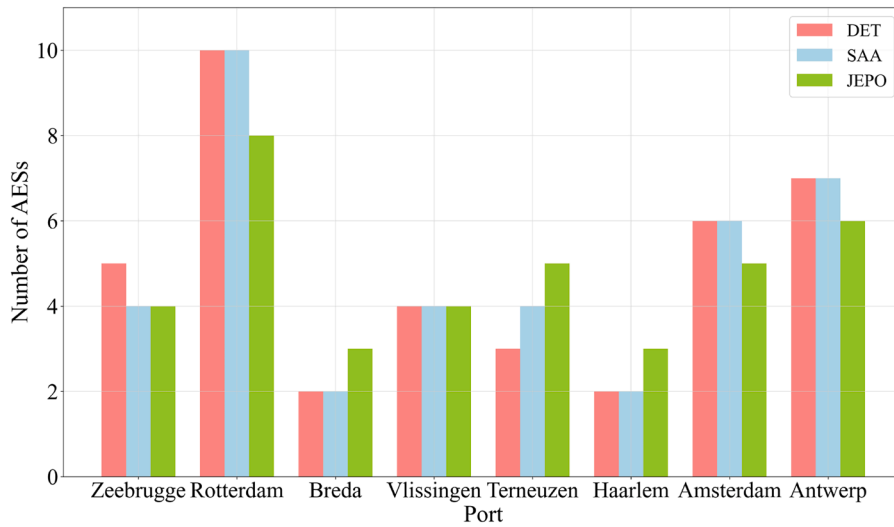


Fig. 5. Initial AESs allocation among 8 ports when $M = 10000$.

Table 2

The computational time of the JEPO model and benchmark models under different M values (seconds).

Budget (M)	DET	SAA	CRO	JEPO
10000	<1	13	72	1764
15000	<1	13	72	1779
20000	<1	14	69	1801
25000	<1	14	75	1961
30000	<1	17	71	2032

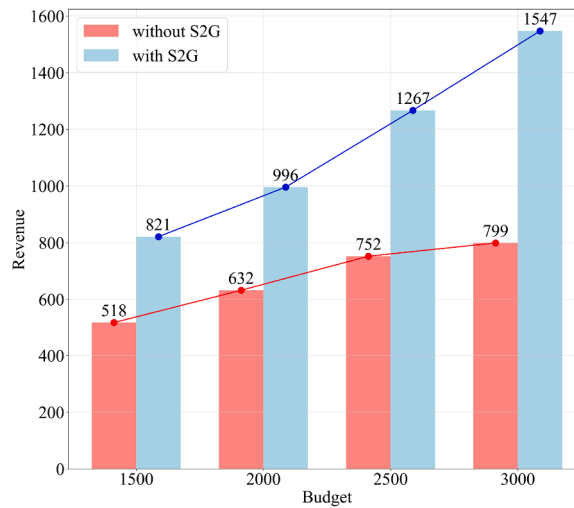


Fig. 6. Revenue with/without S2G under different budgets.

We begin with a small-scale experiment involving three ports. Fig. 6 compares the revenue generated with and without S2G under different budget conditions. The results show that AESs equipped with S2G consistently outperform those without it in terms of revenue across all budget levels. This advantage grows more significant as the budget increases, highlighting the potential of S2G to effectively enhance operational revenue.

We also investigate the impact of seasonal electricity prices on S2G and, consequently, on the profitability of shipping company. As illustrated in Fig. C.2, there are significant differences in daily electricity prices between winter and summer. Specifically, summer electricity prices exhibit relatively stable fluctuations with smaller variance, ranging between 0.85 and 0.98. In contrast, winter

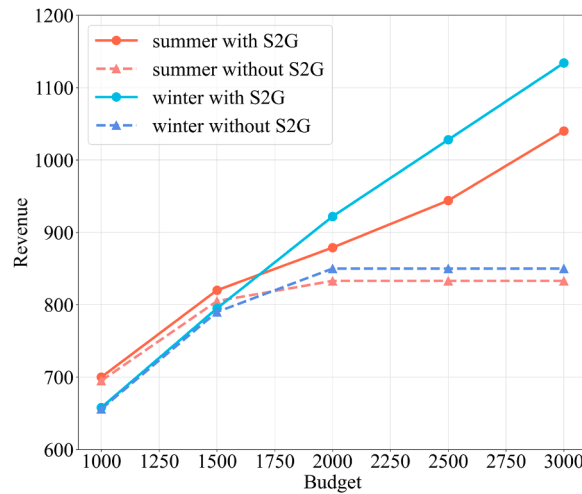


Fig. 7. Benefit of S2G integration and the impact of seasonality.

Table 3

Benefit of S2G integration and the impact of seasonality under $M = 3000$.

Season	With S2G		Without S2G	
	Winter	Summer	Winter	Summer
Revenue				
Freighting	1786	1901	2365	2,203
Discharging	935	574	N/A	N/A
Cost				
Relocating	236	253	412	474
Idle	0	0	231	189
Charging	1293	1179	871	907
Net revenue	1,192	1043	851	833
% increase	40.07	25.21	N/A	N/A

electricity prices show larger fluctuations with greater variance, peaking at 0.95 and dipping as low as 0.79. The seasonal electricity price structure is the core driver of S2G revenue difference.

The larger price differential in winter makes AESs particularly more suitable to operate as flexible “mobile energy storage units”.

Fig. 7 illustrates that the shipping company utilizing S2G consistently achieves higher profits compared to the company that does not, regardless of the season. Furthermore, the profit advantage grows more pronounced as the initial budget increases, due to the larger number of AESs available for charter as the budget expands. Once freight demand is met, idle AESs docked at ports can further enhance revenue by discharging electricity back to the grid.

Fig. 7 also shows a disparity in the benefits of adopting S2G for AESs between winter and summer. Specifically, when the budget is relatively low, the revenue from adopting S2G in summer is significantly higher than in winter. However, as the budget increases, the revenue from S2G adoption in winter exceeds that in summer. This indicates that the revenue-boosting effect of adopting S2G is more pronounced in summer, especially when the budget is relatively low. In winter, as the budget increases, the revenue gains from S2G gradually grow, demonstrating higher revenue potential.

Table 3 shows the detailed of revenue or cost in freighting, discharging, relocating, idle, and charging operations respectively, with or without S2G in winter and summer. The “net revenue” indicates the total profit after accounting for all relevant costs. The final row (“% increase”) represents the percentage of profit improvement achieved with S2G compared with the baseline case without S2G. Several observations can be obtained from Table 3.

First, integrating S2G into AES operations can increase net revenue by generating additional profits from electricity fed back to the power grid. This highlights the substantial economic potential of bidirectional power exchange between AESs and the power system.

Second, the seasonal comparison shows that the financial benefits of S2G are more pronounced in winter. This suggests that the operational value of S2G integration is influenced by both seasonal conditions and the dynamic fluctuations in electricity prices.

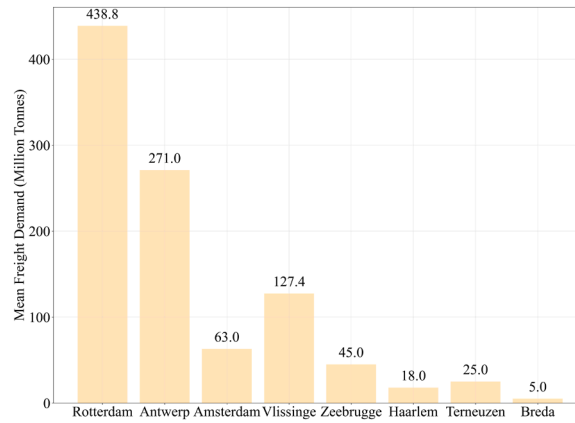


Fig. 8. Average freight demand of 8 selected ports.

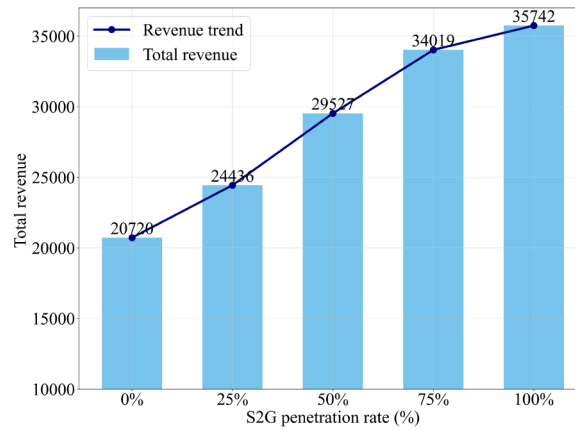


Fig. 9. The impact of S2G penetration on total revenue.

6.4. Case study

In this subsection, we present a case study involving eight representative ports in the North Sea region, Europe, analyzed using our proposed model. These ports include major international hubs such as Rotterdam, Antwerp, and Vlissingen, as well as smaller regional ports, including Breda and Haarlem. Fig. 8 illustrates the average freight demand for each port.

To evaluate the economic benefits under varying S2G levels of adoption, this study performs a sensitivity analysis on the S2G penetration rate within the eight ports under budget $B = 100000$. Here, the S2G penetration rate refers to the proportion of AESs that are capable of bidirectional power exchange with the grid. Five scenarios are considered, corresponding to 0%, 25%, 50%, 75%, and 100% S2G-equipped AESs. All other model parameters, including port electricity prices, charter budget, and freight demand, remain constant to ensure comparability.

Fig. 9 demonstrates that there is a clear upward trend in total revenue as the S2G penetration rate increases. The baseline scenario (0%) yields the lowest revenue, as AESs are unable to discharge electricity to the grid. At 25% penetration, revenue rises by approximately 17.95%. The 50% and 75% scenarios continue this upward trajectory, with profit increases of 14.93% and 15.21%, respectively. However, beyond 75% penetration, marginal gains begin to taper off. Specifically, the 100% scenario provides only a 5.06% improvement compared to 75%. This deceleration in growth suggests that the system is approaching saturation in S2G utilization, as most ports are already equipped with sufficient AESs to discharge electricity back to the power grid.

Fig. 10 presents the eight selected ports (marked as blue) which are equipped with S2G technology. Using the freight demand data among these ports, we apply our model to obtain the distribution and routing of AESs across the North Sea region, Europe. Based on this analysis, we derive the following management insights.

First, the three major hub ports, Rotterdam, Antwerp, and Hamburg, account for over 65% of AESs, as they handle a larger volume of shipping operations. Government should prioritize the deployment of fast-charging/battery-swapping facilities in hub ports and then use feeder services to extend coverage to regional ports like Haarlem.

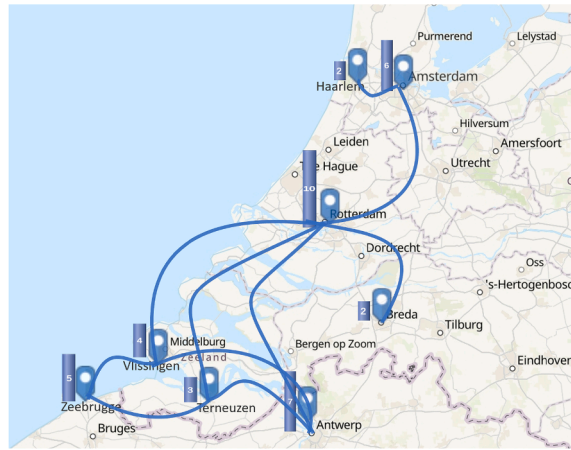


Fig. 10. The distribution and routes of AESs in the North Sea region, Europe.

Second, AESs are concentrated on short-haul routes of less than 200 nautical miles, e.g., Rotterdam–Antwerp, while long-haul routes are constrained by range. To overcome inter-port transshipment bottlenecks, it is urgent to develop “segmented electrification” routes supported by offshore charging infrastructure.

7. Conclusions

Shipping companies face mounting pressure from increasingly stringent environmental regulations and rising fuel costs, making AESs and S2G technology central to the sector’s operational transformation. This study develops a dynamic AES scheduling model on a spatial-temporal-SoC network that captures five key operational behaviors: freight routing, vessel relocation, idle docking, charging, and discharging. We explicitly consider context-dependent uncertainty in AES freight demand, whose fluctuations can be partially inferred from contextual information such as seasonality and hinterland economic indicators. Methodologically, we propose a contextual joint estimation–prediction–optimization (JEPO) framework that establishes a unified data-to-decision paradigm by embedding estimation and prediction procedures directly into the optimization model’s uncertainty set. To reduce excessive conservatism, we further develop a data-driven calibration approach that identifies a minimum-volume confidence ball while preserving the desired confidence level. From a computational perspective, the model is reformulated using linear decision rules and a lifted uncertainty set construction, and a binary-search algorithm with adaptive intervals is employed to ensure convergence to optimal solutions.

We validate the proposed approach using real-world instances from ports in the North Sea region of Europe. The results show that the JEPO model consistently outperforms benchmark models under both average and risk-averse criteria. In addition, comparative experiments indicate that incorporating S2G operations yields further revenue gains, particularly during periods of pronounced electricity price volatility. Sensitivity analyses on S2G penetration further reveal a nonmonotonic effect, with marginal revenue increasing initially and then diminishing as S2G penetration rises. Our findings offer several managerial insights. For shipping companies, profitability can be improved by strategically deploying AESs equipped with S2G capabilities in response to seasonal electricity price variations. At higher levels of S2G penetration, however, firms must carefully balance marginal revenue gains against associated investment and operational costs to maximize overall returns. From a policy perspective, our results highlight the importance of prioritizing battery-swapping infrastructure in major ports and developing *segmented electrification* routes supported by offshore charging facilities to address the challenges of long-haul AES operations.

CRedit authorship contribution statement

Bo Lu: Writing – review & editing, Resources, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization, Validation, Software, Visualization; **Zhuoxi Shi:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis; **Guowei Zhang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Data availability

The authors do not have permission to share data

Appendix A. Notations

Table A.1

List of parameters and variables.

Parameter or variable	Definition
Parameters	
I	The number of ports.
T	The number of time periods.
B	The number of battery states of charge (SoC).
$\mathcal{G} = (\mathcal{N}, \mathcal{A})$	Spatial-temporal-SoC network, $\mathcal{N} \subset [I] \times [T] \times [B]$.
$l_{i,j}$	Sailing time from port $i \in [I]$ to $j \in [I]$.
e^C	Unit of SoC increased during one period of charging.
e^D	Unit of SoC decreased during one period of discharging.
e^F	Unit of SoC decreased during one period of freighting.
e^R	Unit of SoC decreased during one period of relocating.
$r_{i,j}$	Shipping revenue per unit of freight demand from port $i \in [I]$ to $j \in [I]$.
$c_{i,j}^{\text{relo}}$	Cost of relocating an AES from port $i \in [I]$ to $j \in [I]$ per time period.
c_i^{idle}	Cost of an AES idling at port i per time period.
c^{deg}	Unit of battery degradation cost.
p_t^s	Discharging price of one unit electricity at time $t \in [T]$.
p_t^c	Charging price of one unit electricity at time $t \in [T]$.
c_i	Cost of chartering an AES at port $i \in [I]$.
M	Fixed budget.
L	Capacity of an AES.
$\tilde{d}_{i,j,t}$	Amount of freight demand transported from port $i \in [I]$ to port $j \in [I]$ at time $t \in [T]$.
Sets	
$\mathcal{A}^{\text{frgt}}$	Set of freighting arcs.
$\mathcal{A}^{\text{relo}}$	Set of relocating arcs.
$\mathcal{A}^{\text{idle}}$	Set of idle arcs.
$\mathcal{A}^{\text{chrg}}$	Set of charging arcs.
$\mathcal{A}^{\text{dsch}}$	Set of discharging arcs.
$\delta^-(n_{i,t,b})$	Set of arcs of which node $n_{i,t,b}$ is the destination node.
$\delta^+(n_{i,t,b})$	Set of arcs of which node $n_{i,t,b}$ is the origin node.
Decision Variables	
x_i	The number of AESs chartered at the port $i \in [I]$.
y_a^{frgt}	The number of AESs in the freighting arc $a \in \mathcal{A}^{\text{frgt}}$.
y_a^{relo}	The number of AESs in the relocating arc $a \in \mathcal{A}^{\text{relo}}$.
y_a^{idle}	The number of AESs in the idle arc $a \in \mathcal{A}^{\text{idle}}$.
y_a^{chrg}	The number of AESs in the charging arc $a \in \mathcal{A}^{\text{chrg}}$.
y_a^{dsch}	The number of AESs in the discharging arc $a \in \mathcal{A}^{\text{dsch}}$.

Appendix B. Proof of propositions

B.1. Proof of Theorem 1

1. Positive definite. From Definition 1, we have $\rho(\mathbf{d}, \boldsymbol{\zeta}) = \|\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})\|_p$. Since $\mathbf{H}^{-1}$ is a symmetric positive definite matrix, it is invertible, and $\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})$ is a well-defined vector. The L_p -norm $\|\mathbf{x}\|_p$, is always non-negative for any vector $\mathbf{x}$. Thus, we have $\rho(\mathbf{d}, \boldsymbol{\zeta}) \geq 0$, and $\rho(\mathbf{d}, \boldsymbol{\zeta}) = 0$ if and only if $\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta}) = 0$, which implies $\mathbf{d} = \boldsymbol{\zeta}$. Thus, we have shown that $\rho(\mathbf{d}, \boldsymbol{\zeta}) \geq 0$, and $\rho(\mathbf{d}, \boldsymbol{\zeta}) = 0$ if and only if $\mathbf{d} = \boldsymbol{\zeta}$.

2. Convex. We want to prove that for any $\mathbf{d}_1, \mathbf{d}_2 \in \mathcal{D}$ and $\lambda \in [0, 1]$, the following holds: $\rho(\lambda \mathbf{d}_1 + (1 - \lambda) \mathbf{d}_2, \boldsymbol{\zeta}) \leq \lambda \rho(\mathbf{d}_1, \boldsymbol{\zeta}) + (1 - \lambda) \rho(\mathbf{d}_2, \boldsymbol{\zeta})$. From Definition 1, we know $\Delta(\mathbf{d}, \boldsymbol{\zeta}) = \|\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})\|_p$. Since the L_p -norm is a convex function, the composition of the L_p -norm with a linear function, $\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})$ is also convex. Thus, we can conclude that $\rho(\mathbf{d}, \boldsymbol{\zeta})$ is convex.

3. Translation invariant. From Definition 1, we can obtain $\rho(\mathbf{d} - \boldsymbol{\xi}, \boldsymbol{\zeta} - \boldsymbol{\xi}) = \|\mathbf{H}^{-1}((\mathbf{d} - \boldsymbol{\xi}) - (\boldsymbol{\zeta} - \boldsymbol{\xi}))\|_p = \|\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})\|_p = \rho(\mathbf{d}, \boldsymbol{\zeta})$. Thus, we prove that translation invariance holds.

4. Positive homogeneous. From Definition 1, we can obtain $\rho(k\mathbf{d}, k\boldsymbol{\zeta}) = \|\mathbf{H}^{-1}(k\mathbf{d} - k\boldsymbol{\zeta})\|_p = \|k\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})\|_p$. Since $\|k\mathbf{x}\|_p = k\|\mathbf{x}\|_p$ for $k \geq 0$, we have $\Delta(k\mathbf{d}, k\boldsymbol{\zeta}) = k\|\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})\|_p = k\Delta(\mathbf{d}, \boldsymbol{\zeta})$. Thus, we prove that positive homogeneity holds.

5. Symmetric. From Definition 1, we can obtain $\rho(\mathbf{d}, \boldsymbol{\zeta}) = \|\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})\|_p$. Since the L_p -norm is symmetric, meaning $\|\mathbf{x}\|_p = \|\mathbf{x}\|_p$, we have: $\|\mathbf{H}^{-1}(\mathbf{d} - \boldsymbol{\zeta})\|_p = \|\mathbf{H}^{-1}(\boldsymbol{\zeta} - \mathbf{d})\|_p = \rho(\boldsymbol{\zeta}, \mathbf{d})$. Thus, we prove that symmetry holds.

B.2. Proof of Theorem 2

For any $\gamma \geq 0$, we have:

$$\begin{aligned} \mathbb{P}\left\{\mathbf{d} \notin \mathcal{C}(\hat{\mathbf{d}}_0 + \hat{\mathbf{D}}\hat{\mathbf{v}}, \mathbf{H}, \gamma)\right\} &= \mathbb{P}\left\{\|\mathbf{H}^{-1}(\bar{\mathbf{d}} - \hat{\mathbf{d}}^*)\| > \gamma\right\} \\ &\leq \mathbb{P}\left\{\psi(\|\mathbf{H}^{-1}(\bar{\mathbf{d}} - \hat{\mathbf{d}}^*)\|) > \psi(\gamma)\right\} \end{aligned}$$

$$\begin{aligned} &\leq \frac{\mathbb{E}_{\hat{\mathbf{P}}} [\psi(\|\mathbf{H}^{-1}(\tilde{\mathbf{d}} - \hat{\mathbf{d}}^\dagger)\|)]}{\psi(\gamma)} \\ &\leq \frac{1}{\psi(\gamma)}, \end{aligned}$$

where the first inequality is because the deviation penalty function ψ is non-decreasing in $s + \gamma \geq 0$, and the second inequality applies Markov's inequality as ψ is non-negative.

B.3. Proof of Theorem 3

Given the continuity of the deviation penalty function ψ at $r = 0$ and $\psi(0) < 1$, we note that the solution $(q, \mathbf{P}, \mathbf{Q}) = (\mathbf{0}, \mathbf{0}, \epsilon \mathbf{I})$ is feasible for some $\epsilon > 0$. Therefore, the optimal solution must satisfy $\mathbf{Q} \in \mathbb{S}_{++}^{I^2 T}$. Consider the variable substitution $\mathbf{H}^{-1}$ to $\mathbf{Q}$, $\mathbf{H}^{-1} \hat{\mathbf{d}}_0$ to q , and $\mathbf{Q}, \mathbf{H}^{-1} \hat{\mathbf{D}}$ to $\mathbf{P}$. Note that $\mathbf{H} \in \mathbb{S}_{++}^{I^2 T}$ is equivalent to $\mathbf{Q} = \mathbf{H}^{-1} \in \mathbb{S}_{++}^{I^2 T}$. With another variable substitution $\tilde{\mathbf{d}}^\dagger = \mathbf{H}q + \mathbf{H}\mathbf{P}\tilde{\mathbf{v}}$, we then have the following equivalence:

$$\mathbb{E}_{\hat{\mathbf{P}}} [\psi(\|\mathbf{H}^{-1}(\tilde{\mathbf{d}} - \hat{\mathbf{d}}^\dagger)\|)] \leq 1 \iff \mathbb{E}_{\hat{\mathbf{P}}} [\psi(\|\mathbf{Q}\tilde{\mathbf{d}} - q - \mathbf{P}\tilde{\mathbf{v}}\|)] \leq 1.$$

Given that ψ is a convex and non-increasing function, $\mathbb{E}_{\hat{\mathbf{P}}} [\psi(\|\mathbf{Q}\tilde{\mathbf{d}} - q - \mathbf{P}\tilde{\mathbf{v}}\|)]$ is a convex function with respect to the decision variable $(q, \mathbf{P}, \mathbf{Q})$. Hence, $\mathbb{E}_{\hat{\mathbf{P}}} [\psi(\|\mathbf{Q}\tilde{\mathbf{d}} - q - \mathbf{P}\tilde{\mathbf{v}}\|)]$ is a convex constraint.

Consider the case of having a positive regularization vector $\mathbf{r} > \mathbf{0}$. By Schur complement,

$$\mathbf{H} \geq \text{Diag}(\mathbf{r}) \iff \mathbf{H} - \text{Diag}(\mathbf{r}) \geq 0 \iff \begin{bmatrix} \mathbf{H} & \mathbf{I} \\ \mathbf{I} & \text{Diag}^{-1}(\mathbf{r}) \end{bmatrix} \geq 0.$$

By left multiplying with $\begin{bmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{H}^{-1} & \mathbf{I} \end{bmatrix}$ and right multiplying by its transpose, we obtain that

$$\begin{bmatrix} \mathbf{H} & \mathbf{0} \\ \mathbf{0} & \text{Diag}^{-1}(\mathbf{r}) - \mathbf{H}^{-1} \end{bmatrix} \geq 0,$$

which implies that $\text{Diag}^{-1}(\mathbf{r}) \geq \mathbf{H}^{-1} = \mathbf{Q}$. Since $\mathbf{Q}$ is also a positive definite matrix, the constraint $\text{Diag}^{-1}(\mathbf{r}) \geq \mathbf{Q}$ implies $\text{Diag}(\mathbf{r}) \geq \mathbf{Q}^{-1} = \mathbf{H}$. That is to say, the convex constraint $\mathbf{Q} \leq \text{Diag}^{-1}(\mathbf{r})$ after the variable substitution is equivalent to the original constraint $\mathbf{H} \geq \text{Diag}(\mathbf{r})$.

The objective of problem (MVCB) that minimizes $\det(\mathbf{H}) = 1/\det(\mathbf{Q})$, is equivalent to maximizing $\log \det(\mathbf{Q})$, for which the log determinant function is a concave function. This concludes that problem (12) is a convex optimization problem.

B.4. Proof of Theorem 4

Changing decision variable from $\mathbf{H}^{-2}$ to $\mathbf{G}$, model (12) is equivalent to:

$$\begin{aligned} &\sup \quad \log \det(\mathbf{G}) \\ &\text{s.t.} \quad \min_{\hat{\mathbf{d}}_0 \in \mathbb{R}^{I^2 T}, \hat{\mathbf{D}} \in \mathbb{R}^{(N) \times L}} \mathbb{E}_{\hat{\mathbf{P}}} \left[\left\| \mathbf{G}^{1/2}(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0 - \hat{\mathbf{D}}\tilde{\mathbf{v}}) \right\|_2^2 \right] \leq L \\ &\quad \mathbf{G} \in \mathbb{S}_{++}^{I^2 T}. \end{aligned}$$

Note that $\mathbb{E}_{\hat{\mathbf{P}}} \left[\left\| \mathbf{G}^{1/2}(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0 - \hat{\mathbf{D}}\tilde{\mathbf{v}}) \right\|_2^2 \right] = \mathbb{E}_{\hat{\mathbf{P}}} \left[(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0 - \hat{\mathbf{D}}\tilde{\mathbf{v}})^\top \mathbf{G}(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0 - \hat{\mathbf{D}}\tilde{\mathbf{v}}) \right]$ is a quadratic function of $\hat{\mathbf{d}}_0$ and $\hat{\mathbf{D}}$. Since $\mathbf{G} > \mathbf{0}$, equating the gradients with respect to $\hat{\mathbf{d}}_0$ and $\hat{\mathbf{D}}$ then yields $2\mathbb{E}_{\hat{\mathbf{P}}} \left[\mathbf{G}(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0 - \hat{\mathbf{D}}\tilde{\mathbf{v}}) \right] = \mathbf{0} \Rightarrow \hat{\mathbf{d}}_0^* = \hat{\boldsymbol{\mu}}_d - \hat{\mathbf{D}}^* \hat{\boldsymbol{\mu}}_v$, and $-2\mathbb{E}_{\hat{\mathbf{P}}} \left[(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0^* - \hat{\mathbf{D}}^* \tilde{\mathbf{v}}) \tilde{\mathbf{v}}^\top \right] = \mathbf{0} \Rightarrow \hat{\boldsymbol{\Sigma}}_{dv} = \hat{\mathbf{D}}^* \hat{\boldsymbol{\Sigma}}_v$. Hence, $\hat{\mathbf{D}}^* = \hat{\boldsymbol{\Sigma}}_{dv} \hat{\boldsymbol{\Sigma}}_v^{-1}$ and $\hat{\mathbf{d}}_0^* = \hat{\boldsymbol{\mu}}_d - \hat{\mathbf{D}}^* \hat{\boldsymbol{\mu}}_v$. Consequently, we have:

$$\begin{aligned} \mathbb{E}_{\hat{\mathbf{P}}} \left[(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0^* - \hat{\mathbf{D}}^* \tilde{\mathbf{v}})^\top \mathbf{G}(\tilde{\mathbf{d}} - \hat{\mathbf{d}}_0^* - \hat{\mathbf{D}}^* \tilde{\mathbf{v}}) \right] &= \mathbb{E}_{\hat{\mathbf{P}}} \left[\text{tr}(\mathbf{G}(\tilde{\mathbf{d}} - \hat{\boldsymbol{\mu}}_d - \hat{\mathbf{D}}^*(\tilde{\mathbf{v}} - \hat{\boldsymbol{\mu}}_v))(\tilde{\mathbf{d}} - \hat{\boldsymbol{\mu}}_d - \hat{\mathbf{D}}^*(\tilde{\mathbf{v}} - \hat{\boldsymbol{\mu}}_v))^\top) \right] \\ &= \text{tr}(\mathbf{G}(\hat{\boldsymbol{\Sigma}}_d - \hat{\mathbf{D}}^* \hat{\boldsymbol{\Sigma}}_{dv} - \hat{\boldsymbol{\Sigma}}_{dv} (\hat{\mathbf{D}}^*)^\top + \hat{\mathbf{D}}^* \hat{\boldsymbol{\Sigma}}_v (\hat{\mathbf{D}}^*)^\top)) \\ &= \text{tr}(\mathbf{G}(\hat{\boldsymbol{\Sigma}}_d - \hat{\boldsymbol{\Sigma}}_{dv} \hat{\boldsymbol{\Sigma}}_v^{-1} \hat{\boldsymbol{\Sigma}}_{vd})). \end{aligned}$$

Model (12) then becomes as follows:

$$\begin{aligned} &\sup \quad \log \det(\mathbf{G}) \\ &\text{s.t.} \quad \text{tr}(\mathbf{G}(\hat{\boldsymbol{\Sigma}}_d - \hat{\boldsymbol{\Sigma}}_{dv} \hat{\boldsymbol{\Sigma}}_v^{-1} \hat{\boldsymbol{\Sigma}}_{vd})) \leq L \\ &\quad \mathbf{G} \in \mathbb{S}_{++}^{I^2 T}. \end{aligned}$$

The Lagrangian function for this problem is: $\Lambda(\mathbf{G}, \lambda) = \log \det(\mathbf{G}) - \lambda \text{tr}(\mathbf{G}(\hat{\boldsymbol{\Sigma}}_d - \hat{\boldsymbol{\Sigma}}_{dv} \hat{\boldsymbol{\Sigma}}_v^{-1} \hat{\boldsymbol{\Sigma}}_{vd})) + \lambda L$. We can calculate the gradient with respect to $\mathbf{G}$ as $\nabla_{\mathbf{G}} \Lambda(\mathbf{G}, \lambda) = \mathbf{G}^{-1} - \lambda(\hat{\boldsymbol{\Sigma}}_d - \hat{\boldsymbol{\Sigma}}_{dv} \hat{\boldsymbol{\Sigma}}_v^{-1} \hat{\boldsymbol{\Sigma}}_{vd})$. Equating it to zero, the optimal solution shall satisfy: $\lambda^* \mathbf{G}^* = (\hat{\boldsymbol{\Sigma}}_d -$

$\hat{\Sigma}_{dv}\hat{\Sigma}_v^{-1}\hat{\Sigma}_{vd})^{-1}$. Complementary slackness of the KKT conditions requires that $\lambda^* \left(\text{tr} \left(\mathbf{G}(\hat{\Sigma}_d - \hat{\Sigma}_{dv}\hat{\Sigma}_v^{-1}\hat{\Sigma}_{vd}) \right) - L \right) = 0$, implying $\lambda^* = 1$. Therefore, we obtain the optimal solution of the deviation matrix via $\mathbf{G}^* = (\hat{\Sigma}_d - \hat{\Sigma}_{dv}\hat{\Sigma}_v^{-1}\hat{\Sigma}_{vd})^{-1}$, and $\mathbf{H}^* = (\mathbf{G}^*)^{-\frac{1}{2}}$. The optimality is ensured, as problem (12) is a convex optimization problem.

B.5. Proof of Proposition 6

To establish the feasibility of the proposed LDR, we consider problem (16) in its two-stage form. For clarity, we denote the first-stage decision variables as $(\gamma, \mathbf{x}, \mathbf{y})$ and the second-stage decision vector as $\mathbf{k}(\cdot)$. The goal of this proof is to show that for any feasible first-stage decision $(\gamma, \mathbf{x}, \mathbf{y})$, there exists a corresponding second-stage solution $\mathbf{k}(\cdot) \in \mathcal{L}$ that satisfies all second-stage constraints. By establishing that the complete recourse condition holds, we guarantee the existence of a feasible solution for any admissible first-stage decision, thereby confirming the feasibility of the LDR for problem (16).

First, we formulate the corresponding first-stage problem to ensure that the robust budget constraint is satisfied for all demand realizations within the uncertainty set, as follows:

$$\begin{aligned} \max_{\gamma \geq 0} \quad & \gamma \\ \text{s.t.} \quad & (2) - (6) \\ & \sum_{a \in \mathcal{A}^{\text{else}}} c_a y_a - \sum_{i \in [I]} \sum_{j \in [I]} \sum_{t \in [T]} r_{i,j} \left(\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \right) \leq \tau, \quad \forall \tilde{\mathbf{d}} \in \hat{\mathcal{D}}(\gamma, \mathbf{v}). \end{aligned}$$

Given a feasible first-stage decision $(\gamma, \mathbf{x}, \mathbf{y})$ and a realized demand $\tilde{\mathbf{d}} \in \hat{\mathcal{D}}(\gamma, \mathbf{v})$, the second-stage problem can be expressed as:

$$\begin{aligned} \min \quad & k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \\ \text{s.t.} \quad & k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \geq \sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - \tilde{\mathbf{d}}_{i,j,t}(\mathbf{v}) \\ & k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \geq 0 \\ & \mathbf{k}(\cdot) \in \mathcal{L}. \end{aligned}$$

The two scalar constraints can be compactly represented as a single vector inequality. Accordingly, the above problem can be written in matrix form as:

$$\begin{aligned} \min \quad & k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \\ \text{s.t.} \quad & \mathbf{W} k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \geq \begin{bmatrix} \sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - \tilde{\mathbf{d}}_{i,j,t}(\mathbf{v}) \\ 0 \end{bmatrix} \\ & \mathbf{k}(\cdot) \in \mathcal{L}, \end{aligned}$$

where $\mathbf{W} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathbb{R}^{2 \times 1}$ denotes a column vector of ones.

For any vector $\mathbf{h} \in \mathbb{R}^2$, it is straightforward to see that there exists $k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]})$ satisfying $\mathbf{W} k_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \geq \mathbf{h}$. This ensures that the complete recourse condition holds, meaning that for any first-stage decision $\mathbf{y}$, there exists a second-stage decision $\mathbf{k}(\cdot) \in \mathcal{L}$ that satisfies all second-stage constraints.

Specifically, for any given first-stage decisions, we can construct

$$\hat{k}_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]}) \geq \max_{\tilde{\mathbf{d}} \in \hat{\mathcal{D}}(\gamma, \mathbf{v})} \left[\sum_{a \in \mathcal{D}_{i,j,t}^{\text{frgt}}} y_a L - \tilde{\mathbf{d}}_{i,j,t}(\mathbf{v}) \right],$$

and set the LDR decision variables as $k_{0,i,j,t} = \hat{k}_{i,j,t}(\tilde{\mathbf{d}}_{i,j,[t-1]})$, $k_{1,i,j,t,t',j',t'} = 0$.

This provides a feasible solution to the second-stage problem for any admissible first-stage decision. Therefore, problem (16) always admits a feasible solution under the proposed LDR, establishing its feasibility.

B.6. Proof of Proposition 9

Observe that each loop in Algorithm 1 reduces the gap between r_2 and r_1 by half (see Step 3). We now establish the correctness of this binary search. Suppose that $m^r(Z_1) \leq 0$. Then r is feasible in Model (17) and so $r^* \geq r$, since otherwise r would be infeasible in Model (17). Because the function $m^r(Z_1)$ is nondecreasing in r , we have $r^* < r$. Hence, the number of computations K must satisfy: $\bar{r}/2^K < \Delta$; that is, $K \geq \lceil \log_2(\bar{r}/\Delta) \rceil$, which completes the proof.

B.7. Proof of Proposition 4

With Theorem 3, the uncertainty set (11) can be reformulated as:

$$\tilde{D}'(\gamma, \nu) = \left\{ \tilde{\mathbf{d}} \in \mathcal{D} \left| \begin{array}{l} \rho_{\mathbf{H}}(\tilde{\mathbf{d}}, \mathbf{H}\mathbf{q} + \mathbf{H}\mathbf{P}\nu) \leq \gamma \\ \mathbb{E}_{(\mathbf{d}_s, \nu_s) \sim \hat{\mathbb{P}}} [\psi(\|\mathbf{Q}\mathbf{d}_s - \mathbf{q} - \mathbf{P}\nu_s\|)] \leq 1 \\ \det(\mathbf{Q}) \geq 1/\hat{K} \\ \mathbf{Q} \leq \text{Diag}^{-1}(\mathbf{r}) \\ \mathbf{q} \in \mathbb{R}^{I^2T}, \mathbf{P} \in \mathbb{R}^{(N) \times L}, \mathbf{Q} \in \mathbb{S}_{++}^{I^2T} \end{array} \right. \right\}.$$

which is equivalent as follows:

$$\hat{D}(\gamma, \nu) = \left\{ \tilde{\mathbf{d}} \in \mathcal{D} \left| \begin{array}{l} \rho_{\mathbf{H}^*}(\tilde{\mathbf{d}}, \mathbf{H}^*\mathbf{q}^* + \mathbf{H}^*\mathbf{P}^*\nu) \leq \gamma \\ (\mathbf{q}^*, \mathbf{P}^*, \mathbf{Q}^*) = \arg\max \Xi(\mathbf{d}_s, \nu_s), \\ \mathbf{H}^* = (\mathbf{Q}^*)^{-1} \end{array} \right. \right\}$$

where $\Xi(\mathbf{d}_s, \nu_s) = \{\det(\mathbf{Q})^{1/N} | (12b) - (12d)\}$. According to the definition of the contextual disparity metric $\rho_{\mathbf{H}}(\cdot, \cdot)$, the first constraint is equivalent to $\|\mathbf{Q}^*\tilde{\mathbf{d}} - \mathbf{q}^* - \mathbf{P}^*\nu\|_p \leq \gamma$. Therefore, we can obtain:

$$\hat{D}(\gamma, \nu) = \left\{ \tilde{\mathbf{d}} \in \mathcal{D} \left| \begin{array}{l} \|\mathbf{Q}^*\tilde{\mathbf{d}} - \mathbf{q}^* - \mathbf{P}^*\nu\|_p \leq \gamma \\ (\mathbf{q}^*, \mathbf{P}^*, \mathbf{Q}^*) = \arg\max \Xi(\mathbf{d}_s, \nu_s) \end{array} \right. \right\}.$$

B.8. Proof of Proposition 7

Let $\mathbf{r} := \mathbf{Q}^*\tilde{\mathbf{d}} - \mathbf{q}^* - \mathbf{P}^*\nu \in \mathbb{R}^{I^2T}$, we aim to show that $\hat{D}(\gamma, \nu) = D'(\gamma, \nu)$.

(i) $\hat{D}(\gamma, \nu) \subseteq D'(\gamma, \nu)$

Suppose $\tilde{\mathbf{d}} \in \hat{D}(\gamma, \nu)$. By definition, we have $\|\mathbf{r}\|_p = \|\mathbf{Q}^*\tilde{\mathbf{d}} - \mathbf{q}^* - \mathbf{P}^*\nu\|_p \leq \gamma$. We denote:

$$\tilde{\mathbf{w}}_1 := \mathbf{r}^+ := \max(\mathbf{r}, 0), \quad \tilde{\mathbf{w}}_2 := \mathbf{r}^- := \max(-\mathbf{r}, 0), \quad \tilde{u} := \|\tilde{\mathbf{w}}_1 + \tilde{\mathbf{w}}_2\|_p.$$

Clearly, we have $\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2 \geq \mathbf{0}$ and $\tilde{\mathbf{w}}_1 - \tilde{\mathbf{w}}_2 = \mathbf{r}$. Moreover, we can obtain $\|\tilde{\mathbf{w}}_1 + \tilde{\mathbf{w}}_2\|_p = \|\mathbf{r}\|_p = \|\mathbf{r}\|_p \leq \gamma$. Thus, $(\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u})$ satisfies all constraints of $D'(\gamma, \nu)$, and therefore we have $\tilde{\mathbf{d}} \in \hat{D}(\gamma, \nu) \implies (\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \in D'(\gamma, \nu)$.

(ii) $D'(\gamma, \nu) \subseteq \hat{D}(\gamma, \nu)$

Conversely, we suppose $(\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \in D'(\gamma, \nu)$. Then, we have:

$$\mathbf{r} = \tilde{\mathbf{w}}_1 - \tilde{\mathbf{w}}_2, \quad \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2 \geq \mathbf{0}, \quad \|\tilde{\mathbf{w}}_1 + \tilde{\mathbf{w}}_2\|_p \leq \tilde{u} \leq \gamma.$$

For each component, we have $|\mathbf{r}| = |\tilde{\mathbf{w}}_1 - \tilde{\mathbf{w}}_2| \leq \tilde{\mathbf{w}}_1 + \tilde{\mathbf{w}}_2$. Since the p -norm is monotone with respect to componentwise inequalities, we obtain $\|\mathbf{r}\|_p = \|\tilde{\mathbf{w}}_1 - \tilde{\mathbf{w}}_2\|_p \leq \|\tilde{\mathbf{w}}_1 + \tilde{\mathbf{w}}_2\|_p \leq \gamma$. Hence we have $\tilde{\mathbf{d}} \in \hat{D}(\gamma, \nu)$.

We have shown both inclusions $\hat{D}(\gamma, \nu) \subseteq D'(\gamma, \nu)$ and $D'(\gamma, \nu) \subseteq \hat{D}(\gamma, \nu)$. Therefore, we obtain the result of the Proposition 7.

B.9. Proof of Proposition 8

(i) Equivalence of the uncertainty set.

Problem (16) is formulated over the original uncertainty set $\hat{D}(\gamma, \nu)$. Proposition (7) shows that by introducing auxiliary variables $\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}$, the set is an affine lifting of $\hat{D}(\gamma, \nu)$. Hence, for any continuous function f we have

$$\min_{\tilde{\mathbf{d}} \in \hat{D}(\gamma, \nu)} f(\tilde{\mathbf{d}}) = \min_{(\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \in D'(\gamma, \nu)} f(\tilde{\mathbf{d}}),$$

which means optimizing over D' is equivalent to optimizing over $\hat{D}$.

(ii) Affine representation of the LDR in the lifted space.

Under the linear decision rule approximation,

$$k_{i,j,t}(\tilde{\mathbf{d}}) = k_{0,i,j,t} + \sum_{i',j'} \sum_{t'=1}^{t-1} k_{1,i,j,t,i',j',t'} \tilde{\mathbf{d}}_{i',j',t'},$$

which depends only on past demand realizations $t' < t$. In problem (17), it is rewritten as

$$k_{i,j,t}(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) = k_{i,j,t,0} + \mathbf{k}_{i,j,t,1}^\top \tilde{\mathbf{w}}_1 + \mathbf{k}_{i,j,t,2}^\top \tilde{\mathbf{w}}_2 + k_{i,j,t,3} \tilde{u}.$$

To preserve the causal LDR structure, we impose a sparsity pattern on the coefficient vectors:

$$[\mathbf{k}_{i,j,t,1}]_{i',j',t'} = [\mathbf{k}_{i,j,t,2}]_{i',j',t'} = 0 \quad \forall t' \geq t,$$

so that only information from periods $t' < t$ is extracted. This guarantees

$$k_{i,j,t}(\tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) = k_{i,j,t}(\tilde{\mathbf{d}}), \quad \forall (\tilde{\mathbf{d}}, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \tilde{u}) \in D'.$$

The objective function remains unchanged in both formulations, and every robust constraint in Problem (16) can be rigorously transformed into its lifted counterpart in Problem (17) using the affine lifting of the uncertainty set and the LDR representation. Therefore, Problems (16) and (17) define the same feasible region and share the same optimal objective value.

Appendix C. Supplement for figures

C.1. The impact of contextual information on AES freight demand

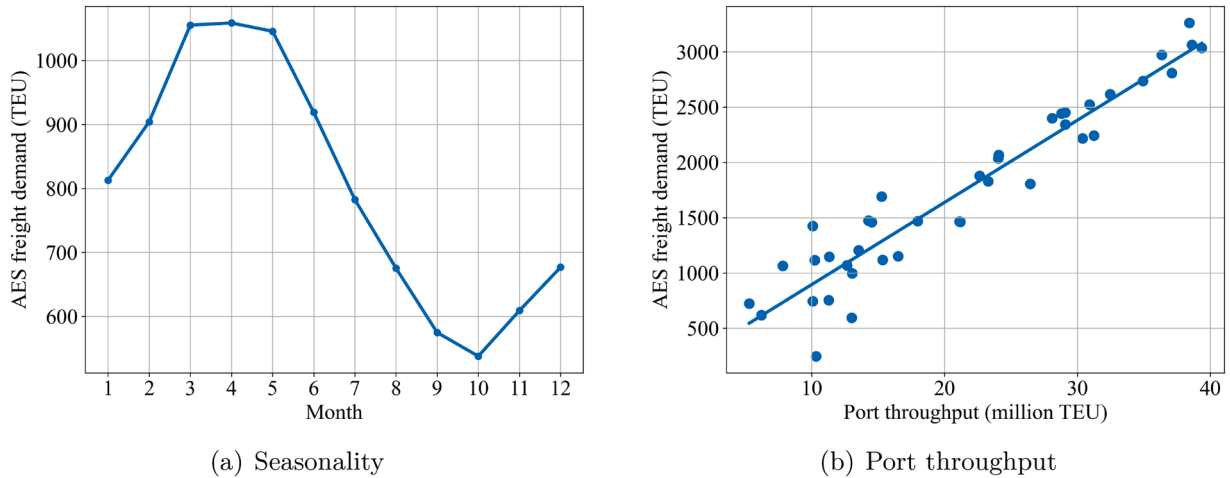


Fig. C.1. The impact of contextual information on AES freight demand.

C.2. The daily electric prices per kWhs in summer and winter

Fig. C.2 illustrates the daily electric prices per kWhs in summer and winter.

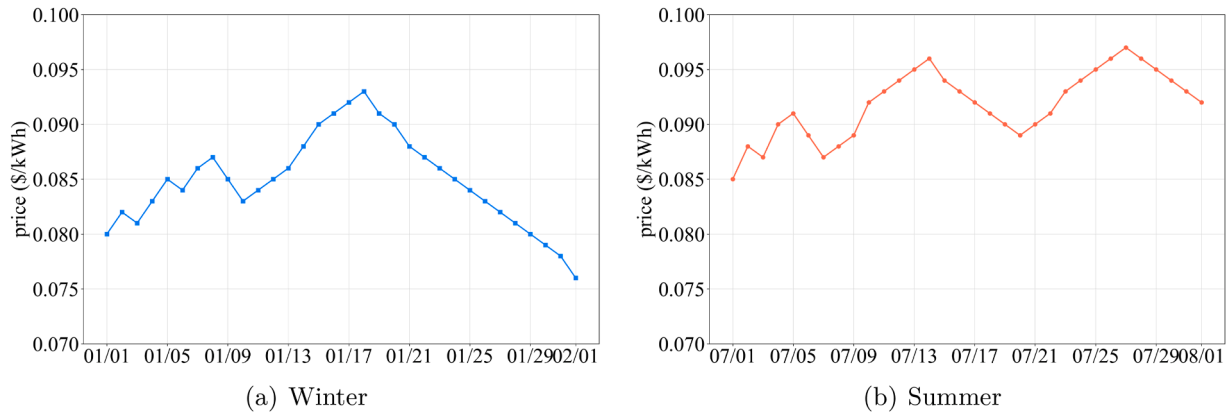


Fig. C.2. Daily electric prices per kWh.

C.3. The 95% CVaR out-of-sample performance under different target levels

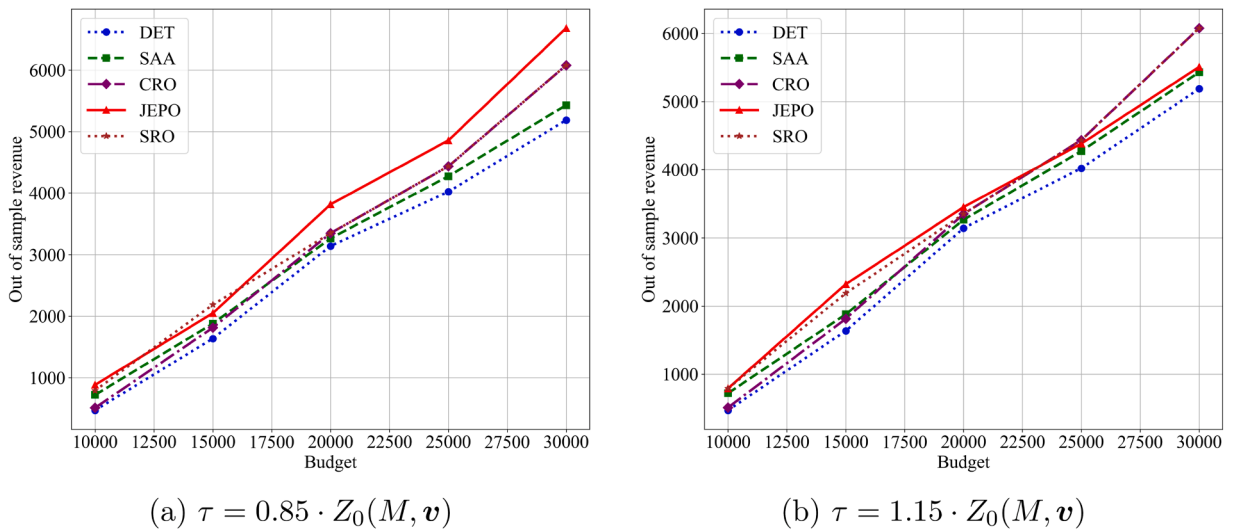


Fig. C.3. 95% CVaR out-of-sample performance under different target levels.. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

References

- Badakhshan, S., Senemmar, S., Zhang, J., 2022. Dynamic feasibility assessment of ship-to-grid interconnection by DC-link. In: 2022 IEEE Texas Power and Energy Conference (TPEC), pp. 1–6.
- Ban, G.-Y., Rudin, C., 2019. The big data newsvendor: practical insights from machine learning. *Oper. Res.* 67 (1), 90–108.
- Bandi, C., Bertsimas, D., 2012. Tractable stochastic analysis in high dimensions via robust optimization. *Math. Program.* 134 (1), 23–70.
- Ben-Tal, A., Den Hertog, D., De Waegenae, A., Melenberg, B., Rennen, G., 2013. Robust solutions of optimization problems affected by uncertain probabilities. *Manag. Sci.* 59 (2), 341–357.
- Bengio, Y., 1997. Using a financial training criterion rather than a prediction criterion. *Int. J. Neural Syst.* 8 (4), 433–443.
- Bertsimas, D., Gupta, V., Kallus, N., 2018. Data-driven robust optimization. *Math. Program.* 167 (2), 235–292.
- Bertsimas, D., Kallus, N., 2020. From predictive to prescriptive analytics. *Manag. Sci.* 66 (3), 1025–1044.
- Bertsimas, D., Sim, M., 2004. The price of robustness. *Oper. Res.* 52 (1), 35–53.
- Bertsimas, D., Sim, M., 2006. Tractable approximations to robust conic optimization problems. *Math. Program.* 107 (1–2), 5–36.
- Brown, D.B., Sim, M., 2009. Satisficing measures for analysis of risky positions. *Manag. Sci.* 55 (1), 71–84.
- Chen, Z., Xiong, P., 2023. Rsome in python: an open-source package for robust stochastic optimization made easy. *INFORMS J. Comput.* 35 (4), 717–724.
- Christiansen, M., Fagerholt, K., Ronen, D., 2004. Ship routing and scheduling: status and perspectives. *Transp. Sci.* 38 (1), 1–18.
- Chuang, F.-C., Lin, C.-H., Lin, C.-S., 2010. A fuzzy linear programming model for robust liner ship fleet planning under uncertain port times. *J. Mar. Sci. Technol.* 18 (4), 583–592.
- DNV, 2024. Maritime Energy Transition Outlook 2024. Technical Report. DNV. Høvik, Norway.
- Donti, P.L., Amos, B., Kolter, J.Z., 2017. Task-based end-to-end model learning in stochastic optimization. In: *Advances in Neural Information Processing Systems (NeurIPS)* 30, pp. 5484–5494.
- Elmachtoub, A.N., Grigas, P., 2022. Smart “predict, then optimize”. *Manag. Sci.* 68 (1), 9–26.
- ESPO, 2025. The European Sea Ports Organisation. [https://www.espo.be/. Online; accessed 12 November 2025].
- EU, 2021. Part of the EU’s 2030 infrastructure strategy supporting alternative fuels infrastructure, including port electrification. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32021R1153>.
- Gürel, S., Krumke, S.O., Qu, R., Voss, S., 2019. Recovery of disrupted liner shipping services by container diversion and vessel swapping. *Transp. Res. E: Logist. Transp. Rev.* 122, 249–270.
- Hao, Z., He, L., Hu, Z., Jiang, J., 2020. Robust vehicle pre-allocation with uncertain covariates. *Prod. Oper. Manag.* 29 (4), 955–972.
- He, L., Hu, Z., Zhang, M., 2020. Robust repositioning for vehicle sharing. *Manuf. Serv. Oper. Manag.* 22 (2), 241–256.
- Hein, K., Xu, Y., Wilson, G., Gupta, A.K., 2021. Coordinated optimal voyage planning and energy management of all-electric ship with hybrid energy storage system. *IEEE Trans. Power Syst.* 36 (3), 2355–2365.
- Hong, L.J., Huang, Z., Lam, H., 2021. Learning-based robust optimization: procedures and statistical guarantees. *Manag. Sci.* 67 (6), 3447–3467.
- IMO, 2023. Fourth IMO Greenhouse Gas Study 2023. Technical Report. International Maritime Organization.
- Ivanov, G., 2022. All-electric cargo ships data analysis and efficiency vs fuel ships comparison. *Int. J. Mar. Eng. Innov. Res.* 7 (1), 33–40.
- Kanellos, F.D., Tsekouras, G.J., Hatziaargyriou, N.D., 2014. Optimal demand-side management and power generation scheduling in an all-electric ship. *IEEE Trans. Sustain. Energy* 5 (4), 1166–1175.
- Kannan, R., Bayraksan, G., Luedtke, J.R., 2024. Residuals-based distributionally robust optimization with covariate information. *Math. Program.* 207 (1–2), 369–425.
- Liyanage, L.H., Shanthikumar, J.G., 2005. A practical inventory control policy using operational statistics. *Oper. Res. Lett.* 33 (4), 341–348.
- Long, D.Z., Sim, M., Zhou, M., 2023. Robust satisficing. *Oper. Res.* 71 (1), 61–82.
- Lou, Z., Chen, Z., Sim, M., Xie, J., Xiong, P., 2024. Estimation and prediction procedures for unified robust decision models. Available at SSRN 4890089.
- Meng, Q., Wang, S., Liu, Z., 2010. Liner shipping service network design with empty container repositioning. *Transp. Res. E: Logist. Transp. Rev.* 46 (6), 960–977.
- Mišić, V.V., Perakis, G., 2020. Data analytics in operations management: a review. *Manuf. Serv. Oper. Manag.* 22 (1), 158–169.
- MOT, 2024. Zero-Carbon Port Initiative Implementation Guidelines. China Communications Press, Beijing.
- Nemhauser, G.L., Wolsey, L.A., 1988. *Integer and Combinatorial Optimization*. Wiley-Interscience Series in Discrete Mathematics and Optimization, John Wiley & Sons, New York.

- Norstad, I., Fagerholt, K., Laporte, G., 2011. Tramp ship routing and scheduling with speed optimization. *Transp. Res. C: Emerg. Technol.* 19 (5), 853–865.
- Omholt-Jensen, S., Fagerholt, K., Meisel, F., 2025. Fleet repositioning in the tramp ship routing and scheduling problem with bunker optimization: a matheuristic solution approach. *Eur. J. Oper. Res.* 321 (1), 88–106.
- Peterson, S.B., Whitacre, J.F., Apt, J., 2010. The economics of using plug-in hybrid electric vehicle battery packs for grid storage. *J. Power Sources* 195 (8), 2377–2384.
- Qi, X., Song, D.-P., Chang, C.-L., 2012. Service network design in liner shipping with empty container repositioning and demand uncertainty. *Transp. Res. B: Methodol.* 46 (10), 1491–1502.
- Reed, G.F., Grainger, B.M., Sparacino, A.R., Mao, Z.-H., 2012. Ship to grid: medium-voltage DC concepts in theory and practice. *IEEE Power Energy Mag.* 10 (6), 70–79.
- Shang, C., Srinivasan, D., Reindl, T., 2016. Economic and environmental generation and voyage scheduling of all-electric ships. *IEEE Trans. Power Syst.* 31 (5), 4087–4096.
- Sim, M., Tang, Q., Zhou, M., Zhu, T., 2024. The analytics of robust satisficing: predict, optimize, satisfy, then fortify. *Oper. Res.* 73 (5), 2708–2728.
- Song, D.-P., Dong, J., Zhang, J., 2015. Liner shipping service network design with schedule reliability. *Transp. Res. B: Methodol.* 71, 1–16.
- Department for Transport, 2021. COP26: Clydebank Declaration for green shipping corridors. GOV.UK. <https://www.gov.uk/government/publications/cop-26-clydebank-declaration-for-green-shipping-corridors> (accessed 12 August 2026).
- Vilhelmsen, C., Lusby, R., Larsen, J., 2014. Tramp ship routing and scheduling with integrated bunker optimization. *EURO J. Transp. Logist.* 3 (2), 143–175.
- Vlachokostas, A., Prousalidis, J., Spathis, D., Nikitas, M., Kourmpelis, T., Dallas, S., Soghomonian, Z., Georgiou, V., 2019. Ship-to-grid integration: environmental mitigation and critical infrastructure resilience. In: 2019 IEEE Electric Ship Technologies Symposium (ESTS), pp. 542–547.
- Wang, S., Chen, Z., Liu, T., 2020. Distributionally robust hub location. *Transp. Sci.* 54 (5), 1189–1210.
- Wang, S., Meng, Q., Liu, Z., 2012. Liner ship fleet planning with stochastic port times and demand. *Transp. Res. E: Logist. Transp. Rev.* 48 (4), 863–880.
- Wen, M., Ropke, S., Petersen, H.L., Larsen, R., Madsen, O.B.G., 2016. Full-shipload tramp ship routing and scheduling with variable speeds. *Comput. Oper. Res.* 70 (Jun.), 1–8.
- Wen, S., Zhao, T., Tang, Y., Xu, Y., Zhu, M., Huang, Y., 2020. A joint photovoltaic-dependent navigation routing and energy storage system sizing scheme for more efficient all-electric ships. *IEEE Trans. Transp. Electr.* 6 (3), 1279–1289.
- WU, 2025. Local Weather Forecast, News and Conditions – Weather Underground. [<https://www.wunderground.com>. Online; accessed 12 November 2025].
- Zhang, Y., Lu, M., Shen, S., 2021. On the values of vehicle-to-grid electricity selling in electric vehicle sharing. *Manuf. Serv. Oper. Manag.* 23 (2), 488–507.